Pricing Illiquidity in Strategic Asset Allocation: A Calibration Framework for Practitioners

Candidate

Lorenzo Meloncelli

Supervisor

Prof. Emilio Barucci

Co-supervisor

Viviana Gisimundo

Quantitative Finance - Chair of Asset Management

Academic Year 2025/2026

Abstract

Standard mean-variance optimization, while elegant and widely adopted by institutional investors, treats liquid and illiquid asset classes symmetrically, ignoring the return drag and portfolio constraints that locking up capital imposes in practice. The penalty cost approach proposed by Hayes, Primbs and Chiquoine (2015) extends the Markowitz framework through a marginal penalty curve that captures portfolio-level liquidity preferences in a flexible and economically intuitive way. However, the identification of this curve is left largely to subjective judgement, limiting its practical applicability.

This thesis, developed in collaboration with Amundi Quant Solutions, addresses this gap by proposing a simulation-based calibration framework for the marginal illiquidity penalty curve. The framework quantifies the return drag that illiquidity imposes on a portfolio, arising from forced asset sales at a discount, secondary market haircuts, and missed rebalancing opportunities, through a Monte Carlo simulation of a portfolio subject to asset-specific lock-up constraints. Simulated return losses are then mapped onto a power penalty function, whose parameters are estimated via nonlinear least squares, yielding a calibrated marginal cost curve grounded in portfolio-level simulation rather than subjective assessment.

The framework is applied to three institutional investor archetypes, a complementary pension fund, a university endowment, and a family office, using Amundi's Capital Market Assumptions. For each archetype, the calibrated penalty curve is used to solve the penalized mean-variance optimization problem, producing efficient frontiers and optimal portfolio allocations that reflect each investor's true liquidity preferences. Results show that the penalty curve materially reshapes the efficient frontier and portfolio composition, reducing allocations to illiquid asset classes relative to the unconstrained case and generating a meaningful surplus for investors who price illiquidity correctly.

Contents

1 Introduction

1.1 The illiquidity challenge in institutional asset allocation

Liquidity, in the context of asset management, refers to the ease and speed with which an investment can be converted into cash without a significant loss of value. A perfectly liquid asset, such as a short-term government bond or a money market instrument, can be sold at any time at a price close to its fair value, with minimal transaction costs and no constraints on exit timing. An illiquid asset, by contrast, is one where exit is constrained: either because there is no active secondary market or because contractual lock-up periods prevent early redemption. Private equity funds, real estate partnerships, and hedge funds with redemption gates are canonical examples.

Illiquid asset classes have become a central component of institutional portfolios over the past two decades. Their appeal is straightforward: by accepting constraints on the timing and ease of exit, investors expect to harvest an illiquidity premium over otherwise comparable liquid investments. According to the Preqin Institutional Allocation Study (2025), this trend has accelerated markedly in recent years. Endowments and foundations allocated a higher share of their assets under management to alternatives than any other institutional investor segment in 2024, while private pensions have actively increased their alternatives allocations between 2020 and 2024, with a particular focus on private equity. For large institutional investors, allocations to illiquid asset classes often represent a substantial fraction of total portfolio value.

Yet the integration of illiquid assets into a strategic asset allocation framework raises a set of challenges that standard portfolio theory is ill-equipped to handle. Capital committed to illiquid investments cannot be freely redeployed in response to changing market conditions, new investment opportunities, or unexpected liquidity needs. Lock-up periods, redemption gates, and the thinness of secondary markets all constrain the portfolio manager's ability to rebalance, meet obligations, or take advantage of tactical opportunities. These constraints impose real costs on the portfolio, costs that are not captured by expected return and volatility alone. The nature of these costs is inherently non-linear: at low levels of illiquidity the constraints are rarely binding and the costs are modest, but as illiquidity increases the probability of facing a binding constraint grows, the set of available responses shrinks, and the cost of each additional unit of illiquidity rises more than proportionally. Crucially, the degree of this convexity is not universal. It depends on the specific investor, their liability structure, payout obligations, and tolerance for portfolio constraints. A pension fund with near-term benefit payments faces a steeper and more rapidly accelerating cost curve than a family office with no contractual obligations, even at identical levels of portfolio illiquidity.

1.2 Existing approaches and their limitations

The academic literature has approached the problem of illiquidity in asset allocation from several directions. One strand proposes fully dynamic, multi-period stochastic programming models, such as those developed in Mulvey, Pauling and Madey (2003), which incorporate detailed cash flow models of illiquid investments and liabilities across multiple periods. While these models can capture the specific illiquidity characteristics of different asset classes with great precision, they tend to generate computationally demanding optimization problems, particularly when many asset classes and long time horizons are involved. More fundamentally, the added numerical complexity comes at the expense of transparency and intuition, two properties that are essential when asset allocation decisions must be communicated to investment committees, board members, and external constituents.

A more tractable strand of the literature attempts to modify the mean-variance framework directly. Lo, Petrov and Wierzbicki (2003) introduce the notion of a mean-variance-liquidity frontier, exploring modifications to the standard optimization that incorporate a portfolio-level liquidity term into the objective function. Kinlaw, Kritzman and Turkington (2013) propose the use of shadow assets and liabilities to capture liquidity effects from rebalancing, market timing, and capital calls. While these contributions represent meaningful advances, they share a common limitation: the marginal cost of illiquidity is treated as constant, independent of how illiquid the portfolio already is. This assumption is at odds with the convex nature of illiquidity costs described above, and it prevents these frameworks from capturing the investor-specific dimension of liquidity preferences that is central to any rigorous approach to illiquid asset allocation.

1.3 The Hayes, Primbs and Chiquoine (2015) framework

Hayes, Primbs and Chiquoine (2015) propose a framework that directly addresses these limitations. Their approach modifies the standard mean-variance objective by subtracting a portfolio-wide accumulated illiquidity penalty, defined as the integral of a marginal penalty curve up to the current portfolio illiquidity level. The marginal penalty curve captures the return premium that an incremental illiquid investment must earn over a theoretically identical liquid counterpart in order to be a competitive portfolio addition, as a function of the portfolio's current illiquidity level. By construction, this curve is increasing in the portfolio illiquidity level, reflecting the convexity of illiquidity costs.

The framework has several attractive properties. It preserves the intuition and familiarity of mean-variance optimization, allowing for a natural, return-based language for discussing illiquidity preferences with investment committees and board members. It accommodates fractional illiquidity scores across asset classes, rather than treating liquidity as a binary characteristic. And it generates economically interpretable quantities, including illiquidity-adjusted expected returns and an illiquidity surplus measure that captures the net benefit of holding illiquid assets. However, the framework leaves one critical question unanswered: how should the marginal penalty curve be identified? The authors acknowledge that constructing the curve requires careful consideration of a fund's objectives, liabilities, and investment opportunities, and that simulation models are typically required to assess the overall costs of illiquidity. In practice, this means the curve is often specified through subjective judgement, calibrated to a few reference points chosen by the investment team. This subjectivity limits the framework's replicability and its credibility as a tool for rigorous institutional decision-making.

1.4 This paper's contribution

This thesis addresses the calibration gap in the Hayes et al. (2015) framework by developing a systematic, simulation-based methodology for identifying the marginal illiquidity penalty curve. The core idea is to estimate the curve empirically, by simulating the return drag that illiquidity imposes on a portfolio across a range of illiquidity levels, and then fitting a parametric penalty function to the simulated costs.

The simulation models a portfolio subject to asset-specific lock-up constraints, a minimum liquidity buffer, and a stochastic annual payout requirement. In each scenario, the portfolio incurs costs from three sources: forced sales of locked assets at a secondary market discount when the liquidity buffer falls below its minimum, haircuts on illiquid positions that must be liquidated to meet payout obligations, and return drag from the inability to rebalance freely to the optimal target weights. The difference in realized return between the constrained portfolio and a frictionless liquid benchmark, averaged across scenarios, yields an estimate of the illiquidity cost at each portfolio illiquidity level. These point estimates are then used to fit a power penalty function, recovering the parameters of the marginal cost curve through nonlinear least squares. A second contribution of this thesis is to connect the shape of the penalty curve to the characteristics of specific investor types, defining three institutional investor archetypes, a complementary pension fund, a university endowment, and a family office, and calibrating the simulation parameters for each. The resulting curves differ meaningfully across archetypes, reflecting genuine differences in liquidity needs and the associated costs of holding illiquid assets.

1.5 Institutional context: Amundi Quant Solutions

This thesis was developed in collaboration with the Quant Solutions team at Amundi Asset Management. The team comprises four professionals and operates within Amundi's Retirement Solutions division, working closely with the Amundi Investment Institute, portfolio managers, and other colleagues across the Solutions division. The team is responsible for modelling expected returns, including the production of Capital Market Assumptions, and for providing asset class simulations and quantitative analyses for institutional client projects in areas such as strategic asset allocation and asset-liability management, and is also actively involved in producing research on retirement-related themes. The Capital Market Assumptions used throughout this analysis, including expected returns, volatilities, and correlations for all asset classes considered, were provided by Amundi Quant Solutions, and the choice of asset universe, illiquidity scores, and archetype definitions was guided by the team's experience in designing strategic asset allocation frameworks for institutional clients across Europe.

1.6 Structure of the paper

The remainder of this thesis is organized as follows. Section 2 presents the theoretical framework of Hayes et al. (2015) in detail. Section 3 describes the asset universe and Capital Market Assumptions. Section 4 discusses the functional forms considered for the penalty curve and motivates the choice of the power specification. Section 5 introduces the three investor archetypes and their subjective penalty curve parametrizations. Section 6 develops the simulation-based calibration methodology and presents the estimated penalty curves for each archetype. Section 7 presents the results of the penalized mean-variance optimization and their portfolio implications. Section 8 concludes and explores potential directions for future research.

2 Theoretical Framework

2.1 Mean-Variance Optimization: setup and notation

The starting point of our framework is the classical mean-variance optimization problem introduced by Markowitz (1952). We consider a portfolio allocated across n risky asset classes. Let $\boldsymbol{w} = (w_1, \ldots, w_n)^T$ denote the vector of portfolio weights, $\mu = (\mu_1, \ldots, \mu_n)^T$ the vector of expected returns, and Σ the $n \times n$ covariance matrix of returns, with σ_{ij} denoting the covariance between asset classes i and j and $\sigma_{ii} = \sigma_i^2$. The standard mean-variance optimization problem is:

$$\max_{\mathbf{w}} \quad \mathbf{w}^T \mu - \frac{\gamma}{2} \mathbf{w}^T \Sigma \mathbf{w} \tag{1}$$

subject to $w^T 1 = 1$ and $w \geq 0$, where γ is the risk aversion coefficient and 1 is the n-vector of ones. The term $w^T \mu$ is the portfolio expected return and $w^T \Sigma w$ is the portfolio variance. By varying γ from high to low, the optimizer traces the efficient frontier from minimum variance to maximum return portfolios.

It is worth noting the behaviour of the objective as $\gamma \to 0$. In this limit, the variance term vanishes and the optimizer seeks only to maximize expected return, concentrating the portfolio in the highest-returning asset class. In the penalized version of the problem introduced below, this limit plays a different role: the optimizer instead maximizes expected return net of the illiquidity penalty, and the solution reflects a balance between return and the cost of illiquidity rather than between return and variance. This implies a natural lower bound on the penalized frontier that depends directly on the aggressiveness of the penalty curve.

2.2 Illiquidity representation

Following Hayes et al. (2015), we assign each asset class i an illiquidity index $l_i \in [0, 1]$, where $l_i = 0$ denotes a perfectly liquid asset class and $l_i = 1$ denotes a completely illiquid one. The assignment of illiquidity scores is at the user's discretion and should reflect the specific liquidity characteristics relevant to the investor. In our implementation, scores are first assigned on a raw scale reflecting qualitative liquidity characteristics, and then normalized to the unit interval as:

$$l_i = \frac{s_i - s_{\min}}{s_{\max} - s_{\min}} \tag{2}$$

where s_i is the raw score of asset class i and s_{min}, s_{max} are the minimum and maximum scores in the universe. We collect the illiquidity indices in the vector $\boldsymbol{l} = (l_1, \ldots, l_n)^T$.

The overall illiquidity of the portfolio is summarized by the portfolio illiquidity level:

$$L(\mathbf{w}) = \mathbf{w}^T \mathbf{l} = \sum_{i=1}^{n} w_i l_i \tag{3}$$

This quantity can be interpreted as the percentage of the portfolio that is illiquid, and is a linear function of the portfolio weights. For example, a portfolio with equal weights in a perfectly liquid asset and a fully illiquid asset has $L = 0.5$. Note that $L(w) = 0.5$ is a portfolio-level quantity: it captures the aggregate illiquidity of the portfolio rather than the illiquidity of any individual position, and it changes continuously as portfolio weights change.

2.3 The marginal penalty function and the accumulated penalty

To incorporate illiquidity costs into the optimization, Hayes et al. (2015) introduce a marginal illiquidity penalty function $c(L)$, defined over the portfolio illiquidity level $L \in [0, 1]$. The value $c(L)$ represents the return premium that the next incremental unit of illiquid investment must earn over a theoretically identical liquid counterpart in order to be a competitive addition to a portfolio currently at illiquidity level L. The key economic property of $c(L)$ is that it is increasing in L: the marginal cost of illiquidity rises as the portfolio becomes more illiquid, reflecting the convex nature of liquidity costs discussed in Section 1.

The total illiquidity cost borne by a portfolio at illiquidity level L is given by the accumulated penalty:

$$T(L(\mathbf{w})) = \int_0^{L(\mathbf{w})} c(x), dx \tag{4}$$

This quantity represents the area under the marginal penalty curve from zero to the current portfolio illiquidity level, and can be interpreted as the total return drag imposed by illiquidity on the portfolio. It is a non-linear function of the portfolio weights through $L(w)$, and it is this non-linearity that introduces the investor-specific dimension into the optimization problem.

2.4 The penalized optimization problem

The penalized mean-variance optimization problem is obtained by subtracting the accumulated penalty from the standard mean-variance objective:

$$\max_{\mathbf{w}} \quad \mathbf{w}^T \mu - \frac{\gamma}{2} \mathbf{w}^T \Sigma \mathbf{w} - T(L(\mathbf{w})) \tag{5}$$

subject to $w^T 1 = 1$ and $w \geq 0$. The penalty term $T(L(w))$ acts as a return deduction that grows with portfolio illiquidity, discouraging excessive concentration in illiquid asset classes without imposing hard constraints. Unlike a fixed allocation cap, the penalty creates a continuous trade-off between the return contribution of illiquid assets and the cost they impose on the portfolio, with the trade-off becoming increasingly unfavourable as illiquidity rises.

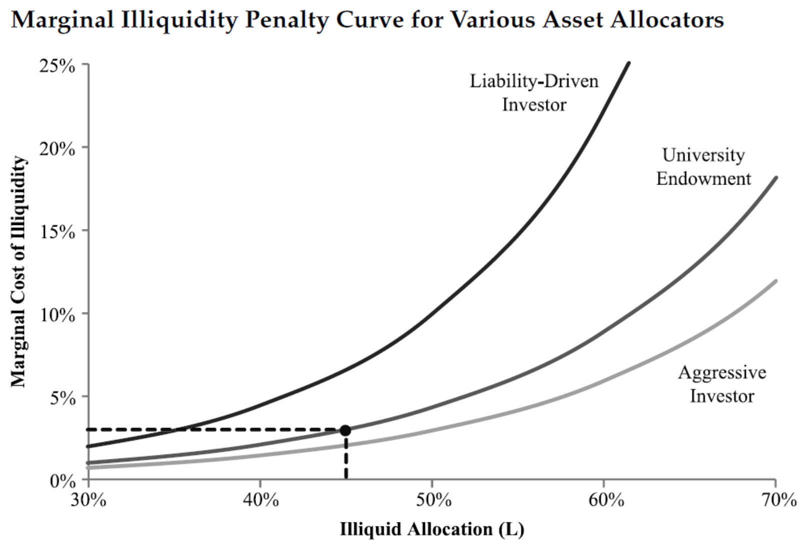

Figure 1: Marginal illiquidity penalty curve for three investor archetypes. The dotted line illustrates that, at a portfolio illiquidity level of 45%, the marginal return penalty for a university endowment is approximately 3%. Source: Hayes, Primbs and Chiquoine (2015).

The non-linearity introduced by the penalty term has important implications for the structure of optimal portfolios. In the standard unpenalized problem, the only trade-off is between expected return and variance, which is a smooth and continuous function of the weights. The addition of the penalty term introduces a third dimension to the trade-off, between return, variance, and illiquidity cost. This third term can create non-linearities in the objective that produce abrupt regime switches in the optimal portfolio composition as the risk aversion coefficient γ varies. In particular, as γ decreases and the optimizer is willing to accept more risk, the penalty may cause sudden shifts in the allocation between liquid and illiquid asset classes, rather than the gradual transitions observed in the unpenalized case.

2.5 Illiquidity-adjusted expected returns

An attractive feature of the penalized formulation is that it can be recast in terms of illiquidity-adjusted expected returns, preserving the familiar mean-variance structure of the optimization. Combining the expected return term and the penalty, the objective can be written as:

$$\mathbf{w}^T\mu - T(L(\mathbf{w})) = \mathbf{w}^T\hat{\mu}(\mathbf{w}) \tag{6}$$

where the vector of illiquidity-adjusted expected returns is defined as:

$$\hat{\mu}(\mathbf{w}) = \mu - \frac{T(L(\mathbf{w}))}{L(\mathbf{w})} \cdot l \tag{7}$$

The quantity $T(L)/L$ is the average accumulated penalty per unit of portfolio illiquidity, and it is applied to each asset class proportionally to its illiquidity index l_i. A perfectly liquid asset with $l_i = 0$ requires no adjustment, while a fully illiquid asset with $l_i = 1$ bears the full average penalty. The penalized problem can therefore be written equivalently as a standard mean-variance optimization with adjusted expected returns:

$$\max_{\mathbf{w}} \quad \mathbf{w}^T \hat{\mu}(\mathbf{w}) - \frac{\gamma}{2} \mathbf{w}^T \Sigma \mathbf{w} \tag{8}$$

subject to $w^T 1 = 1$ and $w \geq 0$. This reformulation is particularly useful for communication purposes: the effect of the illiquidity penalty can be expressed entirely in terms of a downward adjustment to the expected returns of illiquid asset classes, a language that is natural and familiar to practitioners.

2.6 The illiquidity surplus

Given an optimal portfolio w^*, Hayes et al. (2015) define the illiquidity surplus as:

$$S(\mathbf{w}^*) = c(L(\mathbf{w}^*)) \cdot L(\mathbf{w}^*) - T(L(\mathbf{w}^*)) \tag{9}$$

This quantity measures the net benefit that the portfolio derives from holding illiquid assets. At the optimal portfolio, the marginal cost of illiquidity is $c(L(w^*))$, which represents the premium required at the margin to justify an additional unit of illiquid investment. For all previous units of illiquidity up to $L(w^*)$, however, the required premium was lower than $c(L(w^*))$, since the marginal cost curve is increasing. The surplus captures the difference between what the portfolio could in principle demand at the margin and what it actually paid on average, and is geometrically equal to the area above the penalty curve and below the horizontal line at $c(L(w^*))$, up to the illiquidity level $L(w^*)$. A positive illiquidity surplus indicates that the portfolio is being compensated more than adequately for the illiquidity it holds at all but the marginal unit, and it provides a measure of the value created by the decision to invest in illiquid asset classes.

3 Data and Asset Universe

3.1 Asset class universe and Capital Market Assumptions

The empirical analysis is conducted on a universe of six asset classes, chosen to span the full spectrum of liquidity characteristics relevant to a diversified institutional portfolio. The universe includes Money Market USD (MM_US), Global Aggregate Fixed Income hedged in USD (GLOB_AGG), Global Equity USD (GLOB_EQ), Hedge Funds Fund of Funds (HF_FOF), Global Real Estate (GLOB_RE), and Global Private Equity (GLOB_PE). Together, these asset classes cover the main building blocks of a strategic asset allocation, from the most liquid cash-equivalent instruments to long-duration illiquid private market investments.

Expected returns, volatilities, and correlations for all asset classes were provided by the Quant Solutions team at Amundi Asset Management as part of their Capital Market Assumptions framework (Amundi Investment Institute, 2026). The CMA process combines the Amundi CASM simulation model, macroeconomic scenario analysis conducted by the Amundi Investment Institute, and asset class specific adjustments developed by the Quant Solutions team. All figures are based on a starting date of 31 December 2025 and reflect a ten-year forward-looking horizon. Expected returns are expressed in local currency, nominal and gross of fees for liquid asset classes, and net of management and administrative fees for private and alternative assets. The 2026 edition of the CMA is framed around the notion of structural rupture in the global investment environment, and concludes that ten-year expected returns remain broadly attractive across asset classes, with private equity identified as the main growth engine and a key source of return enhancement for institutional portfolios. The figures used in this analysis are summarized in Table 1, alongside the illiquidity scores described in the following section.

3.2 Illiquidity scoring

The illiquidity index l_i for each asset class is constructed through a two-step procedure. In the first step, each asset class is assigned a raw illiquidity score s_i on a qualitative scale that reflects its liquidity characteristics: the ease of exit, the depth and reliability of the secondary market, the typical length of contractual lock-up periods, and the magnitude of transaction costs in stressed conditions. Money market instruments receive the lowest score, reflecting near-instantaneous liquidity at minimal cost. Global aggregate fixed income and global equity are assigned scores close to the minimum, reflecting deep and continuously traded markets. Hedge funds receive an intermediate score, reflecting the presence of redemption gates and notice periods that constrain exit timing. Global real estate and private equity receive progressively higher scores, reflecting long lock-up periods, limited secondary market liquidity, and significant transaction costs.

In the second step, raw scores are normalized to the unit interval using a linear transformation:

$$l_i = \frac{s_i - s_{\min}}{s_{\max} - s_{\min}} \tag{10}$$

where $s_{min} = 1$ and $s_{max} = 5$ are the minimum and maximum scores in the universe. This normalization ensures that MM_US maps exactly to $l_i = 0$ and GLOB_PE maps exactly to $l_i = 1$, while intermediate asset classes receive fractional scores that reflect the continuum of liquidity characteristics across the universe. The use of fractional scores, rather than a binary liquid or illiquid classification, is a key feature of the Hayes et al. (2015) framework, as it allows the optimizer to treat illiquidity as a continuous dimension of each asset class rather than a hard constraint.

3.3 Correlation structure

The correlation matrix of the six asset classes is reported in Table 2. Several features of this matrix are worth noting. Money market and global aggregate fixed income are essentially uncorrelated with all equity-like asset classes, confirming their role as diversifying, liquid components of the portfolio. Global equity, hedge funds, and private equity exhibit very high pairwise correlations, in the range of 0.86 to 0.87, reflecting their common exposure to broad equity market risk. Global real estate is moderately correlated with both the equity cluster and with private equity.

The high correlation between GLOB_EQ, HF_FOF, and GLOB_PE has important implications for the optimization. Since these three asset classes are near-perfect substitutes from a correlation standpoint, the optimizer tends to select only one of them as the representative of the equity risk factor. Given that GLOB_PE offers the highest expected return in the universe at 9.80%, at a volatility broadly comparable to global equity, the unconstrained optimizer concentrates the illiquid allocation almost entirely in private equity, eliminating hedge funds and, to a large extent, global equity from the optimal portfolio. This substitution effect is a key driver of the results discussed in Sections 6 and 7, and it motivates the role of the illiquidity penalty in diversifying the allocation across the equity cluster.

Table 1: Asset class universe: expected real returns, annualised volatilities, and illiquidity indices. Expected returns and volatilities are expressed as annual percentages. Source: Amundi Investment Institute (2026).

Asset class	Expected return	Volatility	Raw score	l_i
MM_US	3.24%	1.82%	1.00	0.00
GLOB_AGG	4.59%	4.39%	1.45	0.11
GLOB_EQ	7.09%	17.01%	1.50	0.13
HF_FOF	5.18%	7.28%	2.50	0.38
GLOB_RE	5.48%	12.51%	3.00	0.50
GLOB_PE	9.80%	20.00%	5.00	1.00

Table 2: Correlation matrix of the six asset classes. Source: Amundi Investment Institute (2026).

	MM_US	GLOB_AGG	GLOB_EQ	HF_FOF	GLOB_RE	GLOB_PE
MM_US	1.000	0.141	−0.009	−0.011	−0.007	−0.012
GLOB_AGG	0.141	1.000	−0.013	−0.035	0.048	−0.064
GLOB_EQ	−0.009	−0.013	1.000	0.860	0.509	0.874
HF_FOF	−0.011	−0.035	0.860	1.000	0.550	0.870
GLOB_RE	−0.007	0.048	0.509	0.550	1.000	0.643
GLOB_PE	−0.012	−0.064	0.874	0.870	0.643	1.000

4 Functional Forms of the Penalty Curve

4.1 Power, Exponential and Quadratic: analytical properties

The Hayes et al. (2015) framework does not prescribe a specific functional form for the marginal penalty curve $c(L)$, leaving the choice to the practitioner. In this thesis we consider three candidate specifications: the power function, the exponential function, and the quadratic function. For each, we derive the corresponding accumulated penalty $T(L)$ analytically. All three specifications are parametrized with $\alpha = 0.05$ and $\beta = 1.6$ for comparison purposes.

The power specification defines the marginal penalty as:

$$c(L) = \alpha \cdot L^{\beta} \tag{11}$$

with accumulated penalty is:

$$T(L) = \frac{\alpha}{\beta + 1} \cdot L^{\beta+1} \tag{12}$$

The exponential specification defines the marginal penalty as:

$$c(L) = \alpha \cdot e^{\beta L} \tag{13}$$

with accumulated penalty:

$$T(L) = \frac{\alpha}{\beta} \left(e^{\beta L} - 1\right) \tag{14}$$

The quadratic specification defines the marginal penalty as:

$$c(L) = \alpha L + \beta L^2 \tag{15}$$

with accumulated penalty:

$$T(L) = \frac{\alpha}{2} L^2 + \frac{\beta}{3} L^3 \tag{16}$$

Figure 2 illustrates the shape of the three specifications for $\alpha = 0.05$ and $\beta = 1.6$. The top panels show the marginal penalty $c(L)$ and the accumulated penalty $T(L)$, while the bottom panels show the average cost $T(L)/L$ and the stability diagnostic $\frac{\mathrm{d}}{\mathrm{d}L}\left[\frac{T(L)}{L}\right]$ discussed in Section 4.3.

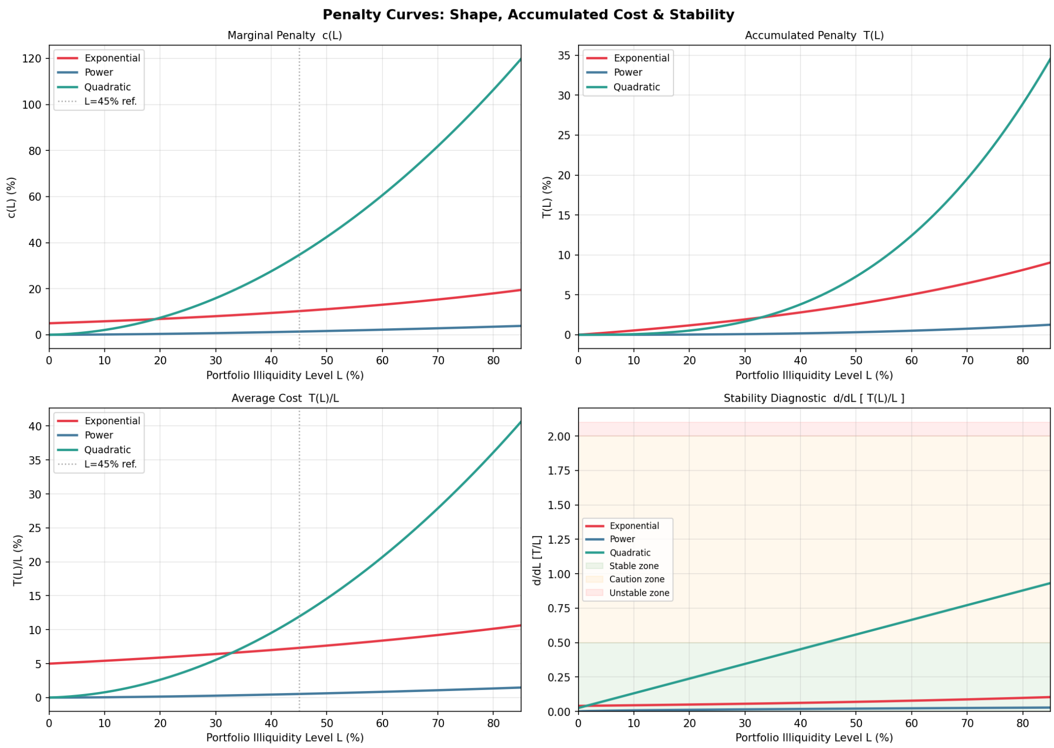

Figure 2: Marginal penalty $c(L)$, accumulated penalty $T(L)$, average cost $T(L)/L$, and stability diagnostic $\frac{\mathrm{d}}{\mathrm{d}L}\left[\frac{T(L)}{L}\right]$ for the three functional forms, parametrized with $\alpha = 0.05$ and $\beta = 1.6$. The dotted vertical line marks $L = 45\%$ as a reference level.

4.2 Economic intuition and desirable properties

Beyond their analytical tractability, the three functional forms differ in important ways from an economic standpoint. A well-behaved penalty curve should satisfy two basic properties: it should equal zero at $L = 0$, reflecting the fact that a fully liquid portfolio bears no illiquidity cost, and it should be strictly increasing and convex in L, reflecting the non-linear nature of illiquidity costs described in Section 1.

Both the power and the quadratic specifications satisfy $c(0) = 0$ by construction, while the exponential does not: at $L = 0$ the exponential penalty equals $\alpha > 0$, implying a strictly positive cost even for a fully liquid portfolio. This is economically unappealing and can distort the optimization by penalizing liquid portfolios that should face no illiquidity cost. The power and quadratic specifications are therefore preferable from a theoretical standpoint.

4.3 Stability diagnostic

A practical concern in the penalized MVO is the numerical stability of the optimization. The illiquidity-adjusted expected return of asset class i is:

$$\hat{\mu}_i(\mathbf{w}) = \mu_i - \frac{T(L(\mathbf{w}))}{L(\mathbf{w})} \cdot l_i \tag{17}$$

The term $T(L)/L$ is the average accumulated penalty per unit of illiquidity, and it enters the adjusted return of every asset class proportionally to its illiquidity index. If this term is highly sensitive to small changes in portfolio weights, the optimization landscape becomes steep and the optimizer may struggle to find stable solutions. The relevant diagnostic is therefore the derivative of $T(L)/L$ with respect to L:

$$\frac{d}{dL}\left[\frac{T(L)}{L}\right] = \frac{c(L) \cdot L - T(L)}{L^2} = \frac{S(L)}{L^2} \tag{18}$$

where $S(L) = c(L) \cdot L - T(L)$ is the illiquidity surplus introduced in Section 2. As shown in the bottom-right panel of Figure 2, the exponential curve generates the largest values of this diagnostic at moderate to high illiquidity levels, confirming its numerical instability. The power and quadratic specifications remain well-behaved across the full range of illiquidity levels considered, with the power curve producing the lowest and most stable values of the diagnostic throughout.

4.4 The effect of the penalty on portfolio composition

The non-linearity introduced by the penalty term has a further implication worth discussing explicitly. In the standard unpenalized MVO, the trade-off is exclusively between expected return and variance, both of which are smooth functions of the portfolio weights. The optimal composition therefore changes continuously as the risk aversion coefficient γ varies, with allocations shifting gradually from lower to higher returning asset classes as γ decreases.

The addition of the penalty term introduces a third dimension to the trade-off. The penalty is a non-linear function of the weights through $L(\mathbf{w})$, and this non-linearity can produce abrupt regime switches in the optimal portfolio composition. As γ decreases and the optimizer becomes willing to accept more risk, the penalty may cause sudden discrete shifts in the allocation between liquid and illiquid asset classes, rather than the smooth transitions observed in the unpenalized case. These regime switches are more pronounced for penalty curves with high curvature, and they tend to occur at illiquidity levels where the marginal cost curve rises steeply.

Figure 3 illustrates this phenomenon clearly. In the unpenalized case, the allocation to GLOB_PE grows gradually and continuously as the target return increases, dominating the portfolio at high return levels. Under the exponential penalty, by contrast, the optimizer essentially eliminates private equity from the portfolio entirely, substituting it with GLOB_EQ and GLOB_AGG, producing an abrupt regime change at low return levels. The power penalty generates a more gradual and economically intuitive transition, preserving a meaningful allocation to private equity at high return targets while reducing it significantly relative to the unpenalized case. The quadratic penalty produces intermediate results, though with a somewhat less smooth transition than the power specification.

The sensitivity of portfolio composition to the choice of parameters is explored systematically in Annex A, which reports the optimal portfolio composition along the penalized frontier for a 6×6 grid of (α, β) values across all three functional forms. The analysis confirms that the power specification produces the most stable and economically intuitive transitions across the parameter space, while the exponential penalty generates abrupt regime switches even at moderate parameter values.

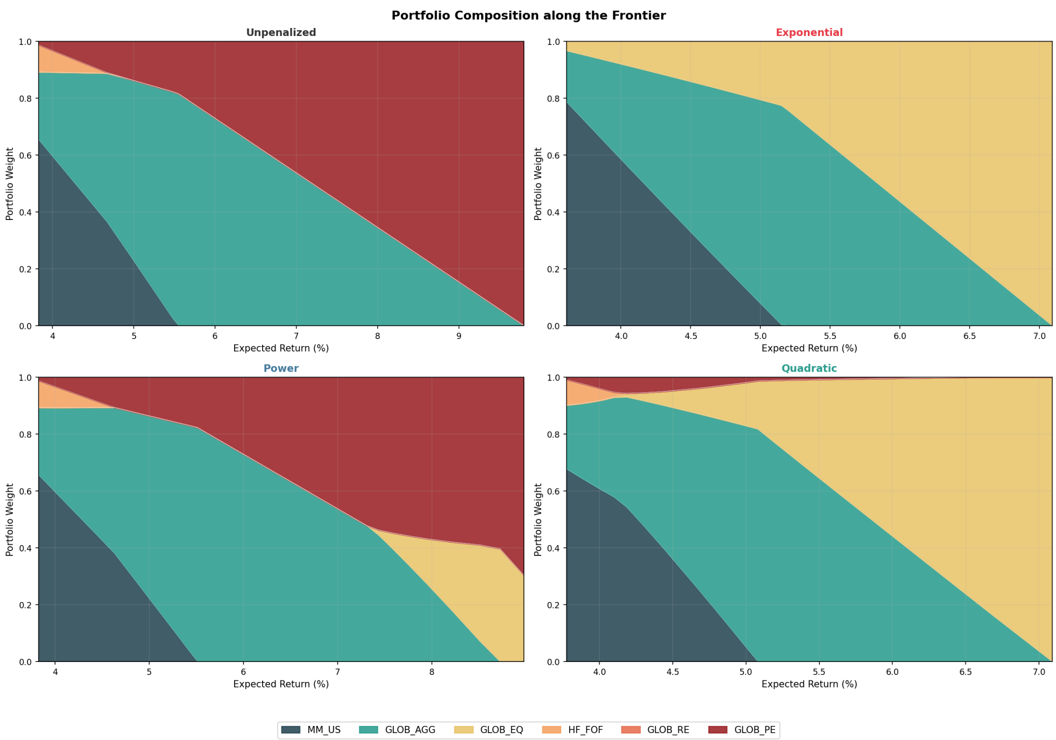

Figure 3: Portfolio composition along the efficient frontier for the unpenalized MVO and the three penalized specifications, parametrized with $\alpha = 0.05$ and $\beta = 1.6$.

4.5 The power function as the reference case

On the basis of the properties discussed above, the power specification is adopted as the reference functional form throughout the remainder of this thesis. It satisfies $c(0) = 0$, it is strictly increasing and convex for $\beta > 0$, it generates sable optimization landscapes, and its two parameters α and β have clear and separable economic interpretations:
The parameter β governs the curvature of the penalty curve and can be interpreted as a measure of how rapidly illiquidity costs accelerate as the portfolio becomes more illiquid. A high β implies that costs are low and rise slowly at moderate illiquidity levels, but accelerate sharply as L approaches high values. This profile is characteristic of investors with long-duration liabilities and stable cash flows, where liquidity constraints are rarely binding until the portfolio is very heavily committed to illiquid assets. A low β, by contrast, implies that costs begin to accumulate meaningfully even at moderate illiquidity levels, reflecting an investor who faces binding liquidity constraints more frequently. In this sense, β is primarily determined by the structural characteristics of the investor's liability side: the duration of liabilities, the openness of the fund, and the stability of cash flows.

The parameter α governs the overall level of the penalty and is more directly linked to market conditions and transaction costs. As $L \to 0$, the penalty $c(L) \to 0$ regardless of α, so α does not represent a fixed cost at zero illiquidity. Rather, it scales the entire curve upward or downward without altering its shape. An investor operating in markets with high secondary market transaction costs, wide bid-ask spreads, or long transaction times will face a higher α than one with access to deep and efficient secondary markets. In this sense, α can be thought of as the baseline illiquidity cost per unit of illiquid exposure, analogous to the level parameter of a cost function: it captures the cost that the investor would face even for a marginal illiquid allocation, before the non-linear effects captured by α begin to dominate.

The practical consequences of identifying the functional form of the penalty with the power function are visible in Figure 4, which shows the liquidity-adjusted net return frontier $\hat{\mu}_P = \mu_P - T(L)$ for all three specifications. The exponential penalty imposes by far the largest return drag across the entire frontier, while the power penalty is the most conservative, reflecting its slower accumulation of costs at moderate illiquidity levels. The gap between the penalized and unpenalized frontiers widens as volatility and illiquidity increase, confirming that the penalty has its largest effect precisely where illiquid allocations are highest.

For the power specification, the penalized MVO problem takes the form:

$$\max_{\mathbf{w}} \quad \mathbf{w}^T \mu - \frac{\gamma}{2} \mathbf{w}^T \Sigma \mathbf{w} - \frac{\alpha}{\beta + 1} \cdot L(\mathbf{w})^{\beta+1} \tag{19}$$

subject to $\mathbf{w}^T \mathbf{1} = 1$ and $\mathbf{w} \geq \mathbf{0}$, where $L(\mathbf{w}) = \mathbf{w}^T \mathbf{l}$. The gradient of the objective with respect to $\mathbf{w}$ is:

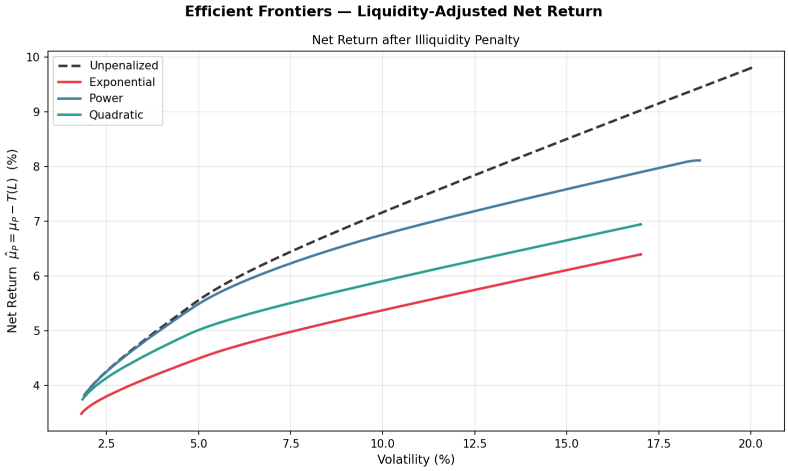

Figure 4: Liquidity-adjusted net return frontier $\hat{\mu}_P = \mu_P - T(L)$ for the unpenalized MVO and the three penalized specifications, parametrized with $\alpha = 0.05$ and $\beta = 1.6$.

$$\nabla_{\mathbf{w}} = \mu - \gamma\Sigma\mathbf{w} - \alpha \cdot L(\mathbf{w})^{\beta} \cdot \mathbf{l} \tag{20}$$

which shows that the penalty enters the first-order conditions as a downward adjustment to the expected return of each asset class, proportional to its illiquidity index l_i and to the marginal cost $c(L) = \alpha \cdot L^{\beta}$ evaluated at the current portfolio illiquidity level. This gradient expression is used directly in the numerical optimization described in Section 6.

5 Investor Archetypes and Penalty Curve Parametrization

5.1 Determinants of the penalty curve shape

The marginal illiquidity penalty curve is not a universal object: its shape depends on the specific characteristics of the investor who holds the portfolio. Two investors with identical asset allocations but different liability structures, payout obligations, and cash flow dynamics will face different illiquidity costs, and should therefore be characterized by different penalty curves. Understanding what drives the shape of the curve is a prerequisite for any systematic calibration methodology.

The most direct determinant is the investor's payout obligation. An investor who must distribute a large and predictable fraction of assets each year faces a high probability of needing to liquidate positions at short notice, and bears a correspondingly high cost when those positions are locked up. The level of the payout obligation governs the overall height of the penalty curve, while its variability and correlation with market conditions determine how steeply the curve rises as illiquidity increases. A fund whose payout demand spikes precisely when markets are falling, making secondary market liquidity thinnest and haircuts largest, faces a particularly convex cost curve.

A second determinant is the maturity structure of existing liabilities and assets. A fund with long-duration liabilities, such as a young pension fund with decades before the bulk of benefits fall due, can afford to lock up capital for extended periods without facing binding liquidity constraints. A fund with short-duration liabilities, by contrast, must maintain a higher liquid buffer and faces a steeper penalty for illiquidity at any given portfolio level. Similarly, the maturity profile of existing illiquid positions matters: a portfolio with a large stock of recently committed capital, all subject to active lock-up periods, faces higher effective illiquidity than one with a more seasoned and diversified vintage structure.

A third determinant is whether the fund is open or closed to new contributions. An open fund with active members, such as a pension fund still receiving contributions or an endowment receiving regular donations, has access to incoming cash flows that can absorb payout obligations without requiring asset sales. This external liquidity buffer reduces the probability of forced selling and flattens the penalty curve. A closed fund, by contrast, must meet all obligations from existing assets, making the liquidity buffer more fragile and the penalty curve steeper.

Finally, the depth and reliability of secondary markets for the fund's specific illiquid holdings matters. A fund concentrated in well-established private equity partnerships has access to a more developed secondary market than one holding bespoke infrastructure debt or direct real estate. The expected haircut in a forced sale, and the time required to complete a transaction, both feed directly into the cost simulation and therefore into the shape of the calibrated penalty curve.

5.2 The three archetypes: Pension Fund, Endowment, Family Office

To make the penalty curve mapping operational, we define three institutional investor archetypes that span the range of liquidity profiles encountered in practice. Each archetype is characterized by a distinct combination of payout dynamics, liability structure, and market access, which translate into a specific set of power curve parameters.

The *pension fund* represents the most liquidity-constrained archetype. It faces mandatory annual distributions to members, including pension payments, early withdrawals, and redemptions, which in aggregate represent a significant and persistent fraction of assets under management. Payout demands tend to be correlated with adverse market conditions, as members are more likely to request early withdrawals during periods of financial stress. The fund is partially open, with active members still contributing, but the contribution flow is insufficient to offset the payout obligation at high illiquidity levels. The combination of high payout, strong market correlation, and regulatory minimum liquidity requirements makes this the archetype with the steepest and most rapidly accelerating penalty curve.

The *university endowment* represents an intermediate liquidity profile. It distributes a stable fraction of assets annually to fund university operations, typically following a smoothing rule that partially decouples the payout from short-term market conditions. The fund is open, receiving regular donations that provide an additional liquidity buffer. Its liabilities are long-duration and relatively predictable, reducing the probability of binding liquidity constraints. The penalty curve is less steep than that of the pension fund, reflecting both the lower payout level and the more stable cash flow dynamics.

The *family office* represents the least liquidity-constrained archetype. It has no contractual payout obligations and distributes assets at the discretion of the beneficiaries. Payout demands are idiosyncratic and weakly correlated with market conditions. The fund is typically closed or semi-closed, with no systematic inflow of new capital, but its absence of binding obligations means that illiquid positions can be held to maturity without penalty in most scenarios. The penalty curve is the flattest of the three, reflecting the low probability of forced selling and the flexibility to time exits optimally.

5.3 Subjective calibration and efficient frontiers

Following the approach of Hayes et al. (2015), each archetype is assigned a set of power curve parameters (α, β) through subjective calibration, based on the qualitative assessment of liquidity needs described above. The complementary pension fund is assigned $\alpha = 0.25$ and $\beta = 1.2$, reflecting its high baseline sensitivity to illiquidity and its relatively uniform cost accumulation across illiquidity levels. The university endowment is assigned $\alpha = 0.12$ and $\beta = 2.0$, reflecting a lower baseline cost and a more convex profile that rises steeply only at high illiquidity levels. The family office is assigned $\alpha = 0.06$ and $\beta = 2.8$, reflecting the lowest baseline sensitivity and the most convex profile, with meaningful costs arising only when the portfolio is very heavily committed to illiquid assets.

These subjective parametrizations serve two purposes. First, they provide a set of benchmark curves against which the simulation-based calibration of Section 6 can be evaluated. Second, they illustrate how the penalty curve framework can be used even in the absence of a full simulation, relying instead on the investment team's qualitative assessment of the fund's liquidity profile. Figure 5 shows the three penalty curves alongside the unpenalized case.

5.4 Portfolio composition across archetypes

The three penalty curves produce meaningfully different efficient frontiers and portfolio compositions, as illustrated in Figures 6 and 7. The pension fund curve, being the steepest, imposes the largest return drag and produces the most conservative frontier: private equity is significantly reduced relative to the

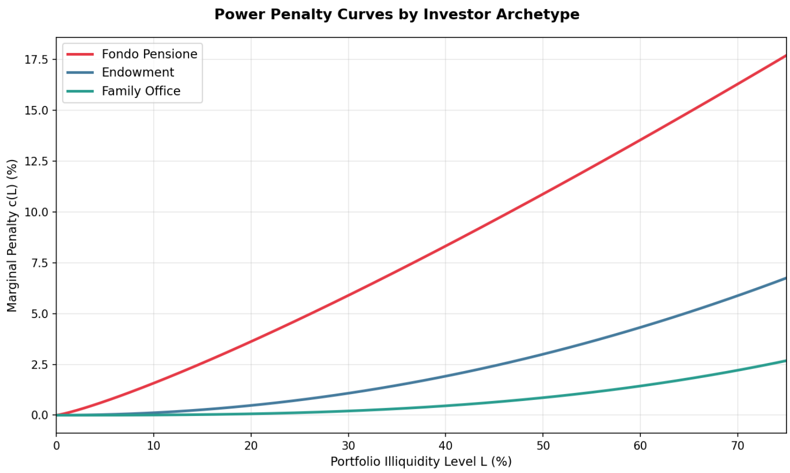

Figure 5: Power penalty curves for the three investor archetypes under subjective calibration. Pension Fund: $\alpha = 0.25$, $\beta = 1.2$. Endowment: $\alpha = 0.12$, $\beta = 2.0$. Family Office: $\alpha = 0.06$, $\beta = 2.8$.

unpenalized case, with the optimizer substituting toward global aggregate fixed income and, to a lesser extent, global equity. The family office curve, being the flattest, produces a frontier closest to the unpenalized case, preserving a larger allocation to private equity across the return spectrum.

A key observation is that the penalty curve does not simply scale down the allocation to all illiquid assets uniformly. Rather, it reshapes the composition of the illiquid bucket, reducing allocations to lower-returning illiquid assets such as real estate more aggressively than to higher-returning ones such as private equity. This reflects the fact that the penalty raises the bar for illiquid investments: only those with sufficiently high expected returns can justify the illiquidity cost they impose on the portfolio. Assets that sit in an intermediate zone, illiquid enough to incur a meaningful penalty but not high-returning enough to compensate for it, are the first to be crowded out by the optimization.

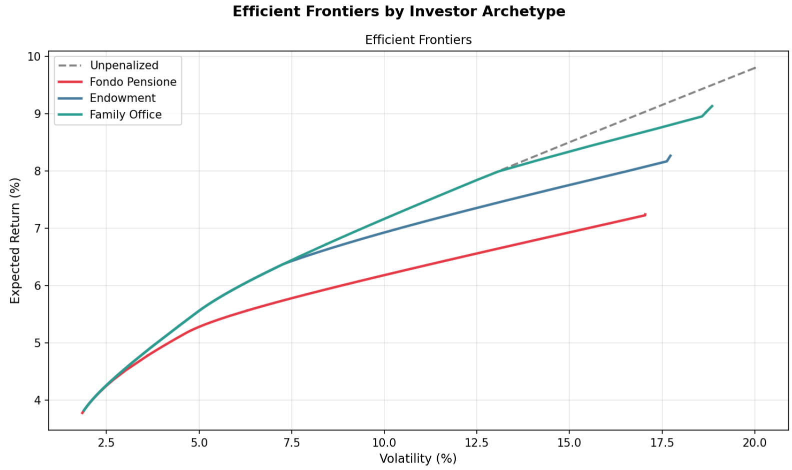

Figure 6: Efficient frontiers for the three investor archetypes under subjective calibration, alongside the unpenalized frontier.

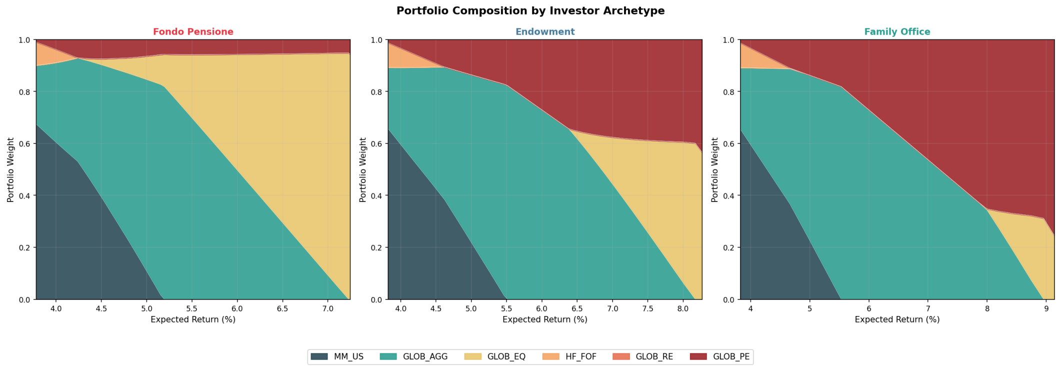

Figure 7: Portfolio composition along the efficient frontier for the three investor archetypes under subjective calibration.

6 Simulation-Based Calibration of the Penalty Curve

6.1 Simulation philosophy: real portfolio versus liquid benchmark

The core idea of the simulation-based calibration is to estimate the illiquidity cost curve empirically, by measuring the return drag that illiquidity imposes on a portfolio across a range of illiquidity levels, and then fitting a parametric penalty function to the simulated costs. The approach is grounded in the observation that the illiquidity penalty curve should reflect what actually happens to a portfolio subject to lock-up constraints, payout obligations, and a minimum liquidity buffer, rather than what an investor believes should happen based on qualitative judgement alone.

The simulation compares two portfolios. The *real portfolio* holds the same target weights as the benchmark but is subject to asset-specific lock-up constraints:

units of an asset purchased at time t cannot be reduced until their lock-up period expires. The *liquid benchmark* holds the same target weights but faces no lock-up constraints: it rebalances freely every year. Both portfolios are subject to identical return shocks, drawn from the same multivariate normal distribution calibrated to the Amundi Capital Market Assumptions, and both pay the same stochastic payout δ_t each year. The benchmark meets the payout obligation by selling any asset at fair value without friction, while the real portfolio may be forced to liquidate locked positions on the secondary market at a discount when liquid assets are insufficient to cover the obligation.

The illiquidity cost in each simulation year is defined as the sum of three components: the return gap between the benchmark and the real portfolio arising from the inability to rebalance freely, the haircut paid on forced sales of locked positions to restore the liquidity buffer, and the haircut paid on locked positions liquidated to meet the payout obligation. Averaging this cost across years and scenarios yields an estimate $\hat{c}(L)$ of the annualised illiquidity cost for a portfolio at illiquidity level L. Running the simulation across a grid of illiquidity levels produces a set of point estimates that trace the shape of the penalty curve.

6.2 Target portfolios

For each illiquidity level L in a grid ranging from 10% to 70%, a target portfolio $\mathbf{w}^*(L)$ is constructed by solving a constrained mean-variance optimization:

$$\max_{\mathbf{w}} \quad \mathbf{w}^T\mu - \frac{\gamma}{2}\mathbf{w}^T\Sigma\mathbf{w} \tag{21}$$

subject to $\mathbf{w}^T\mathbf{1} = 1$, $\mathbf{w} \geq \mathbf{0}$, $\mathbf{w}^T\mathbf{l} = L$, and $\sum_{i\in\mathcal{L}} w_i \geq w^{liq}_{\min}$, where $\mathcal{L}$ denotes the set of liquid asset classes and $w^{liq}_{\min}$ is the minimum liquid fraction required to be held at all times. Both the risk aversion coefficient γ and the minimum liquidity buffer $w^{liq}_{\min}$ are calibrated to each investor archetype, reflecting the differences in risk appetite and regulatory or self-imposed liquidity requirements across investor types. Specifically, the pension fund is assigned $\gamma = 5$ and $w^{liq}_{\min} = 40\%$, consistent with its conservative investment mandate and the regulatory requirement to maintain a significant liquid reserve to meet near-term benefit payments. The endowment is assigned $\gamma = 3$ and $w^{liq}_{\min} = 30\%$, reflecting an intermediate risk tolerance and a liquidity buffer sufficient to cover the annual spending distribution without forced selling in most scenarios. The family office is assigned $\gamma = 2$ and $w^{liq}_{\min} = 20\%$, consistent with its growth-oriented objective and the absence of binding contractual liquidity obligations.

A target portfolio is retained only if it satisfies three validity conditions jointly. First, the realised portfolio illiquidity level must be within two percentage points of the target L. Second, the liquid fraction must meet or exceed $w^{liq}_{\min}$. Third, the portfolio volatility must not exceed 1.5 times the volatility of the preceding valid portfolio in the grid, a filter designed to exclude degenerate solutions in which the optimizer concentrates the portfolio in a single high-volatility illiquid asset class to meet the illiquidity constraint. Portfolios that fail any of these conditions are discarded and the corresponding L level is excluded from the simulation. The composition of the retained target portfolios for each archetype is shown in the right panel of Figures 8, 9, and 10.

6.3 Portfolio accounting and lock-up constraints

The real portfolio is tracked through a system of tranches. Each purchase of an asset is recorded as a separate tranche, identified by the asset class, the weight at the time of purchase, and the year of purchase. A tranche is considered locked if the number of years elapsed since purchase is strictly less than the asset class-specific lock-up period. The lock-up periods and secondary market haircuts used in the simulation are reported in Table 3.

Table 3: Asset-specific lock-up periods and secondary market haircuts used in the simulation.

Asset class	Lock-up (years)	Haircut
MM_US	0	0.1%
GLOB_AGG	0	0.5%
GLOB_EQ	0	2.0%
HF_FOF	2	8.0%
GLOB_RE	5	12.0%
GLOB_PE	8	20.0%

At the start of each simulation, the portfolio is initialised at the target weights with all tranches assigned to year zero, so that no position is locked at inception. In each subsequent year, the simulation proceeds through the following sequence of operations: market returns are drawn and applied to all tranche weights proportionally; the stochastic payout is drawn and distributed across all tranches; if the liquid fraction falls below the minimum buffer $w^{liq}_{\min}$, forced sales are executed to restore it; the illiquidity cost for the year is recorded; and finally the portfolio is rebalanced toward the target weights subject to lock-up constraints.

The rebalancing step solves a constrained quadratic program that minimizes the squared deviation from target weights, subject to the constraint that no locked tranche weight can be reduced and that the liquid fraction remains above $w^{liq}_{\min}$. New purchases are recorded as new tranches with the current year as their vintage.

6.4 Payout, haircuts and liquidity buffer

At each time step, the portfolio must distribute a fraction δ_t of its value to satisfy the payout obligation: the payout is distributed proportionally across all asset classes, regardless of their liquidity status. For tranches that are freely tradeable, the weight is reduced without any additional cost. For locked tranches, the same proportional reduction is applied, but since the asset cannot be sold at fair value, the actual gross amount that must be liquidated exceeds the payout quota by a factor of $(1 - h_i)^{-1}$, where h_i is the secondary market haircut for asset class i reported in Table 3. The difference between the gross amount liquidated and the net proceeds is recorded as a payout haircut cost.

After the payout, the liquid fraction of the portfolio is checked against the minimum buffer $w_{\min}^{liq}$. If the liquid fraction falls below this threshold, the portfolio executes forced sales to restore it. The restoration procedure follows a priority rule: freely tradeable illiquid positions are sold first, without haircut; if these are insufficient, locked illiquid positions are sold with the corresponding secondary market haircut. The proceeds from forced sales are allocated to liquid asset classes in proportion to their target weights. The haircut paid during forced sales is recorded as a liquidity buffer restoration cost. The minimum liquidity buffer therefore plays a dual role in the simulation: it governs the composition of the target portfolios through the constrained MVO, and it determines the frequency and magnitude of forced sales during the simulation, both of which feed directly into the estimated illiquidity cost.

6.5 Stochastic payout process: motivation and parametrization

A central feature of the simulation is that the annual payout rate δ_t is not fixed but follows a stochastic process. This reflects the observation that, in practice, the payout demands of institutional investors are neither constant nor independent of market conditions. A pension fund facing adverse returns is more likely to receive early withdrawal requests from members; a university endowment may be called upon to increase its spending distribution when the institution faces budgetary shortfalls; and a family office may alter its payout in response to the needs of its beneficiaries. The correlation between payout shocks and market returns is therefore a structural feature of the investor's liquidity profile, and its omission would lead to an underestimate of the true illiquidity cost, particularly at high illiquidity levels where the two effects compound.

The payout process is specified as an AR(1) model correlated with the standardised return shock of GLOB_EQ:

$$\delta_t = \mathrm{clip}\Big(\bar{\delta} + \rho_{\mathrm{AR}}\,(\delta_{t-1} - \bar{\delta}) + \eta_t,\ \delta_{\min},\ \delta_{\max}\Big) \tag{22}$$

where the innovation term η_t is defined as:

$$\eta_t = \sigma_\delta\Big(\rho\, z_t^{\mathrm{EQ}} + \sqrt{1-\rho^2}\,\varepsilon_t\Big) \tag{23}$$

with $z_t^{\mathrm{EQ}} \sim \mathcal{N}(0,1)$ the standardised independent shock drawn for GLOB_EQ, $\varepsilon_t \sim \mathcal{N}(0,1)$ an independent idiosyncratic shock, $\sigma_\delta > 0$ the payout volatility, and $\rho \in [-1, 1]$ the target correlation between the payout innovation and the equity market shock..

The form of η_t is chosen to satisfy two properties simultaneously. First, since z_t^{EQ} and ε_t are independent standard normals, the variance of the term in parentheses is:

$$\mathrm{Var}\Big(\rho\, z_t^{\mathrm{EQ}} + \sqrt{1-\rho^2}\,\varepsilon_t\Big) = \rho^2 + (1-\rho^2) = 1 \tag{24}$$

so that $\eta_t \sim \mathcal{N}(0, \sigma_\delta^2)$ exactly, with σ_δ being the standard deviation of the payout innovation regardless of the value of ρ. Second, the correlation between η_t and z_t^{EQ} is exactly ρ:

$$\mathrm{Corr}(\eta_t,\, z_t^{\mathrm{EQ}}) = \frac{\mathrm{Cov}(\sigma_\delta \rho\, z_t^{\mathrm{EQ}},\, z_t^{\mathrm{EQ}})}{\sigma_\delta \cdot 1} = \rho \tag{25}$$

This construction is the bivariate Cholesky decomposition: it builds a normal random variable with prescribed variance σ_δ^2 and prescribed correlation ρ with z_t^{EQ}, without distorting either property as ρ varies.

The parameter ρ_{AR} controls the persistence of payout shocks: a high value implies that an elevated payout in one year is likely to be followed by elevated payouts in subsequent years, reflecting the clustered nature of pension retirements or endowment spending pressures. The parameter ρ controls the direction and strength of the co-movement between payout and market conditions: a negative value, as used for all three archetypes, implies that payouts tend to increase precisely when equity markets fall, amplifying the liquidity stress at the worst possible moment.

The parameters of the payout process are calibrated separately for each investor archetype, drawing on the empirical literature and the qualitative characterisation of Section 5.3. The calibration is summarised in Table 4. For the complementary pension fund, $\bar{\delta} = 10\%$ is consistent with aggregate distribution rates observed in Italian complementary pension funds (COVIP, 2022), which include pension payments, anticipations, and redemptions. The persistence parameter $\rho_{\mathrm{AR}} = 0.65$ reflects the demographic nature of outflows, which tend to concentrate in multi-year retirement waves with a half-life of approximately two years. The correlation $\rho = -0.60$ captures the behavioural tendency of fund members to request early redemptions during periods of market stress, as documented by Dyck and Pomorski (2011) in the context of defined benefit funds. For the university endowment, $\bar{\delta} = 5\%$ follows the canonical value documented in the NACUBO and TIAA Study of Endowments (2019–2023), with lower persistence ($\rho_{\mathrm{AR}} = 0.55$) and weaker equity correlation ($\rho = -0.35$) reflecting the partial decoupling of distributions from short-term market conditions induced by multi-year smoothing rules. For the family office, $\bar{\delta} = 4\%$ is consistent with wealth-preservation-oriented distribution policies (Ang, Papanikolaou and Westerfield, 2013), with low persistence ($\rho_{\mathrm{AR}} = 0.35$) and weak equity correlation ($\rho = -0.20$) reflecting the predominantly idiosyncratic nature of beneficiary liquidity demands (Anson, 2010).

It should be noted that the parameters ρ and ρ_{AR} are not estimated econometrically from the cited literature, but are calibrated for consistency with the empirical patterns documented therein. A direct econometric estimation

Table 4: Payout process parameters by investor archetype.

Archetype	$\bar{\delta}$	σ_δ	ρ_{AR}	ρ	$\delta_{\min}$	$\delta_{\max}$
Pension Fund	10%	2.5%	0.65	−0.60	3%	18%
Endowment	5%	1.5%	0.55	−0.35	2%	9%
Family Office	4%	2.0%	0.35	−0.20	0%	10%

would require time series of archetype-specific payout rates regressed on equity returns, and represents a natural extension of the present analysis.

6.6 Estimating the simulated cost curve

For each target portfolio $\mathbf{w}^*(L)$ and each archetype, the simulation is run over $N = 1{,}000$ independent scenarios of $T = 20$ years. In each year t, a vector of independent standard normal shocks $\mathbf{z}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ is drawn. Asset class returns are then computed as:

$$\mathbf{r}_t = \mu + \mathbf{C}\mathbf{z}_t \tag{26}$$

where $\mathbf{C}$ is the lower Cholesky factor of Σ, so that $\mathbf{r}_t \sim \mathcal{N}(\mu, \Sigma)$. The standardised shock z_t^{EQ}, the component of $\mathbf{z}_t$ corresponding to GLOB_EQ, is then passed directly to the payout process of equation (24), so that the co-movement between payout innovations and equity returns is imposed by construction.

The annualised illiquidity cost for scenario s at illiquidity level L is defined as the time-average of the annual costs:

$$\hat{c}_s(L) = \frac{1}{T}\sum_{t=1}^{T}\left[(r_t^{bench} - r_t^{real}) + \mathrm{HC}_t^{buffer} + \mathrm{HC}_t^{payout}\right] \tag{27}$$

where r_t^{bench} is the return of the liquid benchmark in year t, r_t^{real} is the return of the real portfolio computed on the tranche weights prior to rebalancing, HC_t^{buffer} is the haircut paid to restore the liquidity buffer, and HC_t^{payout} is the haircut paid on locked positions liquidated to meet the payout obligation. The point estimate of the illiquidity cost at level L is then the cross-sectional average:

$$\hat{c}(L) = \frac{1}{N}\sum_{s=1}^{N}\hat{c}_s(L) \tag{28}$$

The benchmark return in year t is computed on the post-drift weights of a frictionless portfolio that holds the same target weights and rebalances freely each period, ensuring that the cost measure captures the drag attributable to lock-up constraints and forced sales, net of the return effect of market drift alone. The real portfolio return is computed before the rebalancing step, so that the rebalancing cost is included in the measurement rather than neutralised by the optimization.

It is important to note the economic interpretation of $\hat{c}(L)$. This quantity is a *total* portfolio cost: it measures the average annual return drag that a portfolio held at illiquidity level L bears relative to its frictionless counterpart, summing across all sources of cost. It is not a marginal cost in the sense of Hayes et al. (2015), and it cannot be used directly as the marginal penalty curve $c(L)$ in the penalized MVO. Rather, $\hat{c}(L)$ corresponds to the ratio $T(L)/L$ in the framework of Section 2: the average accumulated penalty per unit of portfolio illiquidity. This observation motivates the fitting procedure described in the following section.

6.7 Fitting the penalty function and parameter recovery

The simulation produces a set of point estimates $\{(L_k,\, \hat{c}(L_k))\}$ for $k = 1, \ldots, K$. Since $\hat{c}(L)$ estimates the average accumulated penalty $T(L)/L$, the corresponding estimates of the accumulated penalty are:

$$\hat{T}(L_k) = \hat{c}(L_k) \cdot L_k \tag{29}$$

We fit a power function to these accumulated penalty estimates:

$$T(L) = A \cdot L^B \tag{30}$$

by nonlinear least squares, minimising $\sum_k (\hat{T}(L_k) - A \cdot L_k^B)^2$ over (A, B) with constraints $A > 0$ and $B > 1$. The power specification is a natural choice here for two reasons. First, as discussed in Section 4, it satisfies all the desirable analytical properties of a penalty function. Second, and more importantly, the integral of a power marginal cost curve $c(L) = \alpha \cdot L^\beta$ is itself a power function:

$$T(L) = \int_0^L c(x)\, dx = \frac{\alpha}{\beta + 1} \cdot L^{\beta+1} \tag{31}$$

which means that fitting $T(L) = A \cdot L^B$ directly to the simulated accumulated penalties is equivalent to fitting a power marginal cost curve. The parameters of the marginal cost curve are then recovered analytically from the fitted coefficients:

$$\beta = B - 1, \qquad \alpha = A \cdot B \tag{32}$$

so that $c(L) = \alpha \cdot L^\beta$ and $T(L) = \frac{\alpha}{\beta+1} \cdot L^{\beta+1}$, consistent with the power specification of Section 4. This recovery is exact by construction: the constraint $A = \alpha/(\beta + 1)$ is satisfied identically. The lower bound $B > 1$ ensures that the recovered marginal cost $c(L) = \alpha \cdot L^\beta$ is strictly increasing in L, consistent with the convexity requirement of Section 4. The quality of the fit is assessed by the coefficient of determination R^2 computed on the $\hat{T}$ values.

6.8 Results across archetypes

The simulation is run independently for each of the three investor archetypes, using the payout parameters of Table 4, the asset-specific lock-up and haircut parameters of Table 3, and the archetype-specific risk aversion and minimum

Table 5: Risk aversion coefficient and minimum liquidity buffer by investor archetype, used in both the target portfolio construction and the simulation.

Archetype	γ	$w^{liq}_{\min}$
Pension Fund	5	40%
Endowment	3	30%
Family Office	2	20%

Table 6: Fitted penalty function parameters by investor archetype. A and B are the nonlinear least squares estimates of $T(L) = A \cdot L^B$; $\alpha = AB$ and $\beta = B - 1$ are the recovered parameters of the marginal cost curve $c(L) = \alpha \cdot L^\beta$.

Archetype	A	B	α	β	R^2
Pension Fund	0.0359	3.6388	0.1305	2.6388	0.9961
Endowment	0.0288	5.1577	0.1484	4.1577	0.9870
Family Office	0.0055	2.7313	0.0150	1.7313	0.9970

liquidity buffer reported in Table 5. The results are summarised in Table 6 and illustrated in Figures 8, 9, and 10.
The fit quality is high across all three archetypes, with R^2 values ranging from 0.987 to 0.997, confirming that the power function provides an adequate description of the simulated cost curve over the range of illiquidity levels considered.

The results reveal a clear and economically intuitive ordering: the pension fund bears the highest costs at any given illiquidity level, the endowment presents the most convex profile with costs that are negligible at low illiquidity levels but accelerate sharply above a threshold, and the family office exhibits the mildest and most gradually accumulating penalty. This ordering reflects directly the differences in payout demand, market correlation, risk aversion, and minimum liquidity buffer calibrated for each archetype, and is consistent with the qualitative ranking established in Section 5. The most notable divergence from the subjective parametrization of Section 5 concerns the curvature parameter: the simulated β values are systematically higher than their subjective counterparts for the pension fund and endowment, indicating that the simulation assigns a more convex cost profile than qualitative judgement alone would suggest, particularly at high illiquidity levels where forced selling becomes frequent and the interaction between payout shocks and lock-up constraints amplifies the cost nonlinearly.

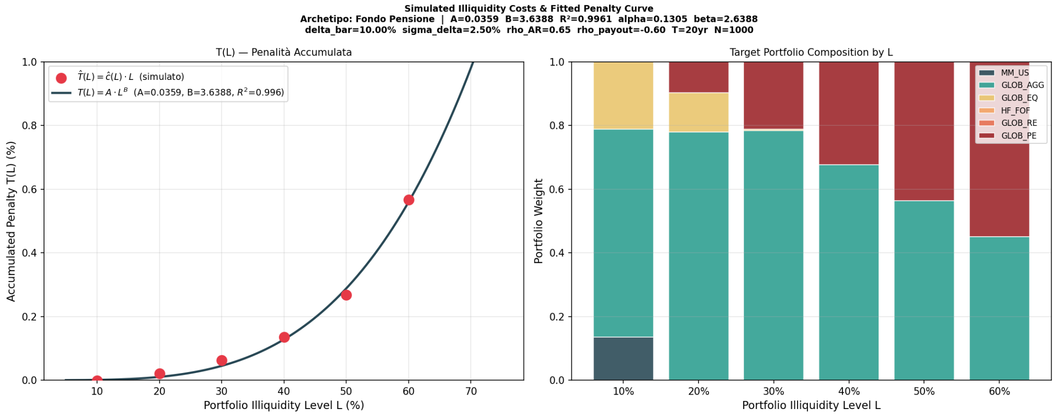

Figure 8: Simulated illiquidity costs and fitted penalty curve for the Pension Fund archetype ($\gamma = 5$, $w_{\min}^{liq} = 40\%$). Left panel: accumulated penalty $\hat{T}(L) = \hat{c}(L) \cdot L$ at each illiquidity level (dots) and fitted power function $T(L) = 0.0359 \cdot L^{3.6388}$ ($R^2 = 0.996$). Right panel: composition of the target portfolios used in the simulation at each illiquidity level. Simulation parameters: $\bar{\delta} = 10\%$, $\sigma_\delta = 2.5\%$, $\rho_{\text{AR}} = 0.65$, $\rho = -0.60$, $T = 20$ years, $N = 1{,}000$ scenarios.

7 Penalized MVO: Results and Portfolio Implications

7.1 Penalized versus unpenalized efficient frontiers

The simulation-calibrated penalty curves derived in Section 6 are used as inputs to the penalized mean-variance optimization of Section 2.4. For each archetype, the resulting penalized frontier is compared to the unpenalized frontier obtained by solving the standard Markowitz problem on the same asset universe and Capital Market Assumptions. Figures 11, 12, and 13 report both frontiers in two representations: the left panel shows the efficient frontier in the standard volatility-return space, while the right panel shows the net return frontier $\hat{\mu}_P = \mu_P - T(L)$, obtained by subtracting the accumulated illiquidity penalty from the gross expected return.

In "gross" return space, the penalized frontier lies below the unpenalized one at high volatility levels and caps out at a lower maximum return, reflecting the fact that the penalty discourages the optimizer from concentrating the portfolio in private equity beyond the point where the marginal illiquidity cost exceeds the return benefit. The shift is not parallel: at low volatility levels the two frontiers coincide, since both optimizers allocate predominantly to liquid asset classes where $T(L)$ is negligible, and the divergence widens progressively as the target return and illiquidity level increase.

The net return frontier provides the economically relevant comparison. It measures what an investor actually expects to retain after accounting for the return drag of illiquidity, and it is in this representation that the penalized frontier dominates the unpenalized one across all three archetypes. An investor

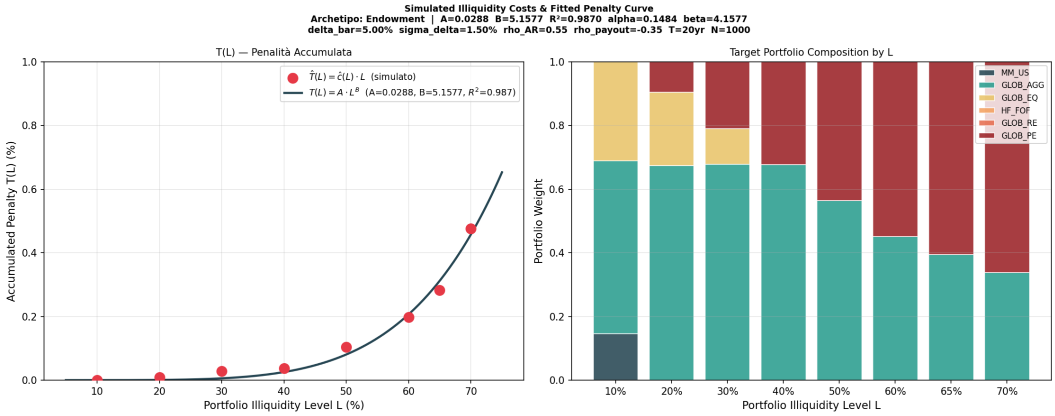

Figure 9: Simulated illiquidity costs and fitted penalty curve for the Endowment archetype ($\gamma = 3$, $w_{\min}^{liq} = 30\%$). Left panel: accumulated penalty $\hat{T}(L)$ (dots) and fitted power function $T(L) = 0.0288 \cdot L^{5.1577}$ ($R^2 = 0.987$). Right panel: composition of the target portfolios at each illiquidity level. Simulation parameters: $\bar{\delta} = 5\%$, $\sigma_\delta = 1.5\%$, $\rho_{\text{AR}} = 0.55$, $\rho = -0.35$, $T = 20$ years, $N = 1{,}000$ scenarios.

who ignores illiquidity costs builds portfolios with high gross returns but also high illiquidity levels, incurring a large accumulated penalty $T(L)$ that erodes the net return. The unpenalized net return frontier therefore turns hump-shaped and eventually declines at high volatility levels, while the penalized net return frontier remains monotonically increasing or only mildly concave. The dominance of the penalized frontier in net return space is the central economic result of the framework: it quantifies the value created by pricing illiquidity correctly.

The magnitude of this dominance differs across archetypes in line with the calibrated penalty parameters. The pension fund and endowment exhibit the largest gap: for the pension fund the unpenalized net return peaks around 12% volatility and then declines sharply to approximately 6.5%, while the penalized frontier delivers above 8% net return at the same point, with a gap exceeding 150 basis points. For the endowment the divergence is concentrated in the high-volatility region, consistent with the highly convex penalty ($\beta = 4.1577$) that is negligible below a threshold and then accelerates rapidly. The family office presents frontiers that are nearly indistinguishable in both representations, consistent with its very low $\alpha = 0.0150$: for this archetype the unpenalized MVO provides a close approximation of the optimal allocation in net return terms.

7.2 Portfolio composition along the frontier

The effect of the penalty curve on portfolio composition is illustrated in Figures 14, 15, and 16. In all three cases, the unpenalized composition follows the same pattern: as the target return increases, the optimizer progressively substitutes GLOB_AGG and MM_US with GLOB_PE, which dominates the portfolio

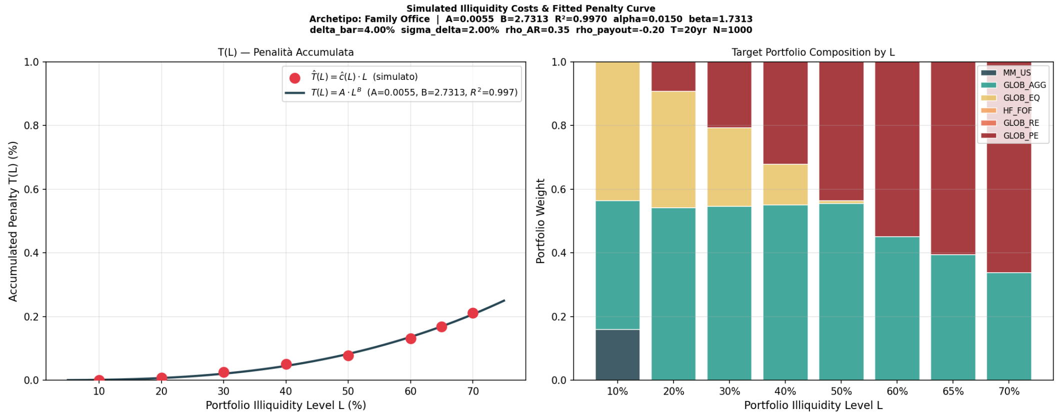

Figure 10: Simulated illiquidity costs and fitted penalty curve for the Family Office archetype ($\gamma = 2$, $w_{\min}^{liq} = 20\%$). Left panel: accumulated penalty $\hat{T}(L)$ (dots) and fitted power function $T(L) = 0.0055 \cdot L^{2.7313}$ ($R^2 = 0.997$). Right panel: composition of the target portfolios at each illiquidity level. Simulation parameters: $\bar{\delta} = 4\%$, $\sigma_\delta = 2.0\%$, $\rho_{\text{AR}} = 0.35$, $\rho = -0.20$, $T = 20$ years, $N = 1{,}000$ scenarios.

at high return levels due to its superior expected return of 9.80% and its high correlation with the other equity asset classes, making it the preferred representative of the equity risk factor. The introduction of the penalty curve materially reshapes this picture, though the degree of reshaping varies across archetypes in line with the aggressiveness of the calibrated curve. The penalty does not eliminate private equity from the optimal portfolio: its return advantage over liquid alternatives is too large for that. Rather, it raises the hurdle rate for illiquid investments, creating space for GLOB_EQ to play a larger role as a liquid substitute for the equity risk factor at intermediate return targets, and caps the maximum attainable return below the unpenalized frontier at the point where the marginal illiquidity cost of adding further private equity exceeds the return benefit.

The simulation-calibrated compositions differ in an important way from those produced by the subjective parametrizations of Section 5.5. The simulated curves have systematically higher β for the pension fund and endowment — $\beta = 2.64$ versus 1.2 and $\beta = 4.16$ versus 2.0 respectively — which implies a penalty that is less aggressive at moderate illiquidity levels but accelerates more sharply at high ones. As a result, the penalized compositions under the simulated curves are closer to the unpenalized case at intermediate return targets and more conservative only at the high end of the frontier, while the subjective calibration produces a broader and more gradual reduction in private equity allocations across the entire return spectrum. For the family office the two calibrations are closer in spirit, though the simulated $\alpha = 0.015$ is substantially lower than the subjective $\alpha = 0.06$, producing compositions that are nearly indistinguishable from the unpenalized case throughout.

These patterns are visible across the three archetypes. The pension fund

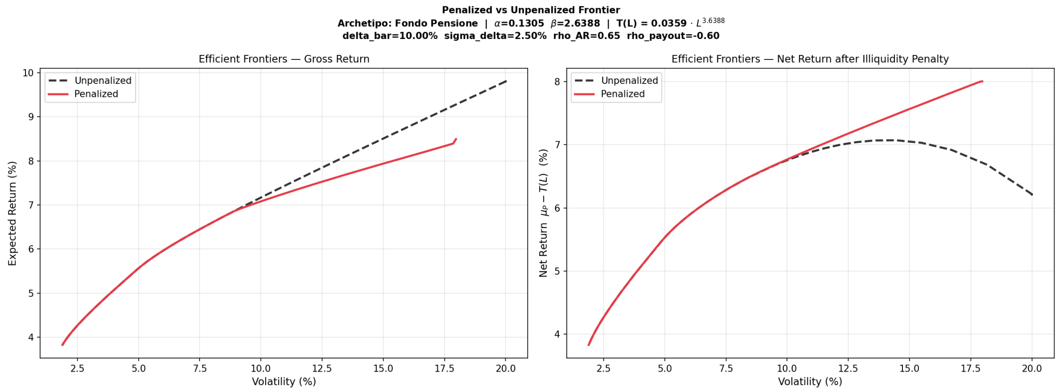

Figure 11: Penalized versus unpenalized efficient frontier for the Pension Fund archetype ($\alpha = 0.1305$, $\beta = 2.6388$). Left panel: gross return frontier. Right panel: net return frontier $\hat{\mu}_P = \mu_P - T(L)$.

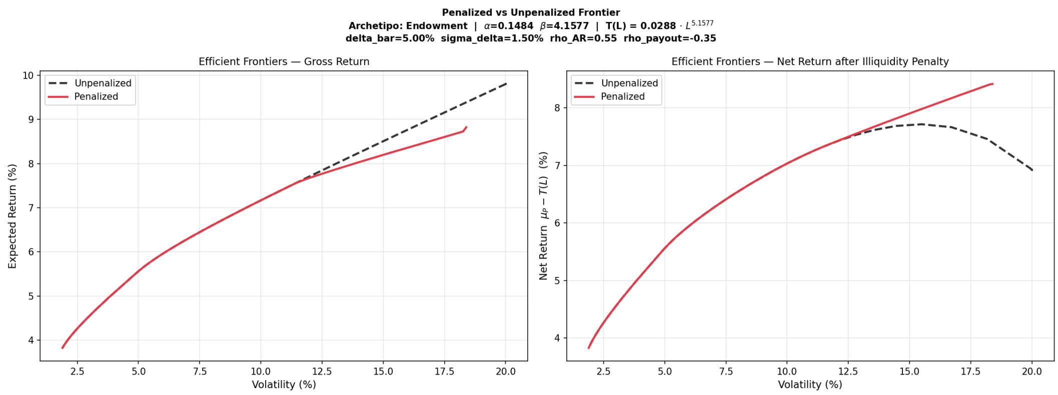

Figure 12: Penalized versus unpenalized efficient frontier for the Endowment archetype ($\alpha = 0.1484$, $\beta = 4.1577$). Left panel: gross return frontier. Right panel: net return frontier $\hat{\mu}_P = \mu_P - T(L)$.

penalized composition shows the most pronounced reshaping: GLOB_EQ emerges as a meaningful component at high return levels, absorbing part of the equity exposure that would otherwise go to GLOB_PE, and the maximum attainable return caps at approximately 8.5%. The endowment displays a distinctive pattern driven by its high β: below the threshold illiquidity level the penalized and unpenalized compositions are nearly identical, but above it GLOB_EQ appears abruptly as a significant component, displacing GLOB_PE in a way that reflects the sudden acceleration of the convex penalty. The family office composition is the closest to the unpenalized case, with GLOB_PE dominating at high return levels in both scenarios and only a marginal increase in GLOB_AGG at intermediate return targets, consistent with the small magnitude of the calibrated penalty.

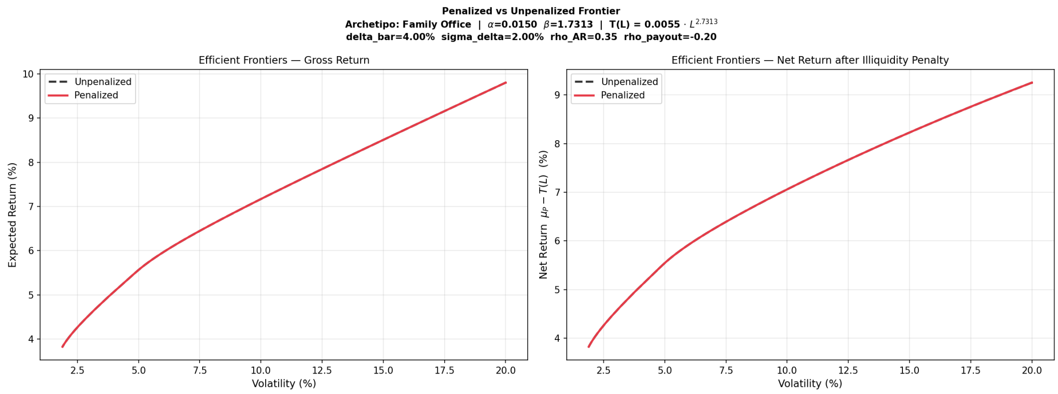

Figure 13: Penalized versus unpenalized efficient frontier for the Family Office archetype ($\alpha = 0.0150$, $\beta = 1.7313$). Left panel: gross return frontier. Right panel: net return frontier $\hat{\mu}_P = \mu_P - T(L)$.

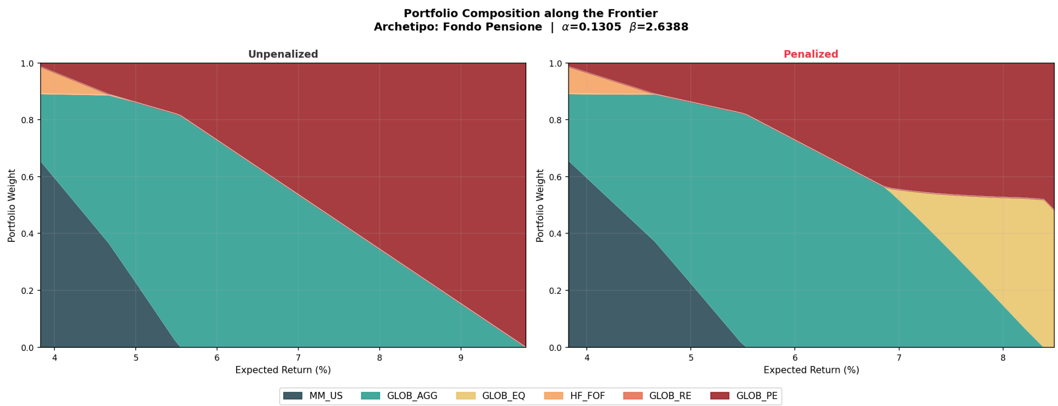

Figure 14: Portfolio composition along the penalized and unpenalized efficient frontier for the Pension Fund archetype ($\alpha = 0.1305$, $\beta = 2.6388$).

8 Conclusions

8.1 Summary of contributions

This thesis has addressed a central gap in the practical application of the Hayes, Primbs and Chiquoine (2015) penalty cost framework for strategic asset allocation with illiquid asset classes: the identification of the marginal illiquidity penalty curve. While the framework provides an elegant and economically intuitive extension of the Markowitz mean-variance optimization, its original formulation leaves the specification of the penalty curve to subjective judgement, limiting its replicability and its credibility as a tool for rigorous institutional decision-making.

The first contribution of this thesis is a simulation-based calibration methodology that grounds the penalty curve in portfolio-level simulation rather than qualitative assessment. The methodology models a portfolio subject to asset-specific lock-up constraints, a minimum liquidity buffer, and a stochastic payout

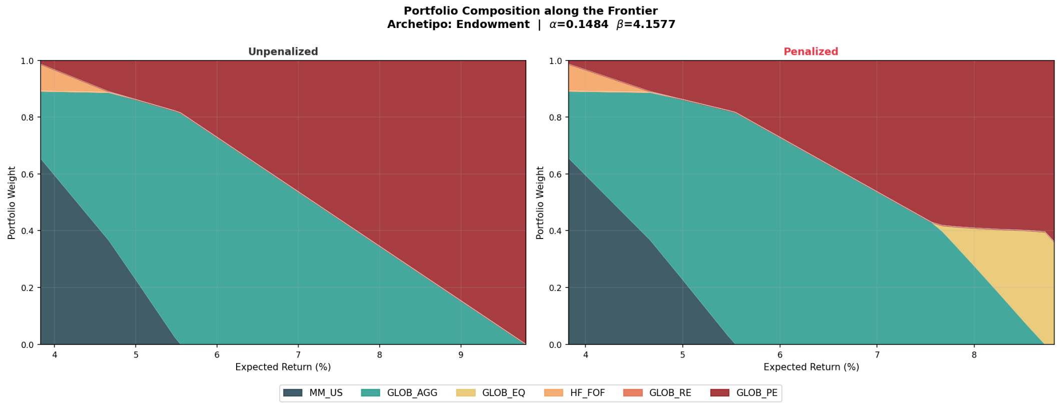

Figure 15: Portfolio composition along the penalized and unpenalized efficient frontier for the Endowment archetype ($\alpha = 0.1484$, $\beta = 4.1577$).

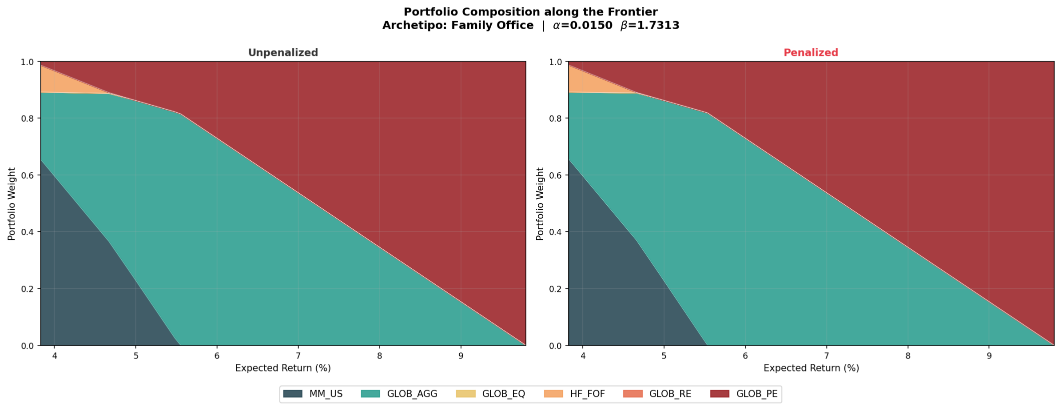

Figure 16: Portfolio composition along the penalized and unpenalized efficient frontier for the Family Office archetype ($\alpha = 0.0150$, $\beta = 1.7313$).

obligation, and measures the return drag that illiquidity imposes relative to a frictionless liquid benchmark across a grid of illiquidity levels. The simulated cost estimates are then used to fit a power penalty function, recovering the parameters of the marginal cost curve through nonlinear least squares. The power specification is shown to be analytically convenient, economically well-behaved, and numerically stable, and the fit quality is high across all three archetypes considered, with R^2 values ranging from 0.987 to 0.997.

The second contribution is a systematic mapping between investor archetypes and penalty curve parameters. Three institutional investor types are defined, a complementary pension fund, a university endowment, and a family office, each characterised by a distinct payout process, liquidity buffer, and risk aversion coefficient. The simulation produces calibrated penalty curves that differ meaningfully across archetypes in both level and curvature, reflecting genuine differences in liquidity needs and the associated costs of holding illiquid assets. The calibrated curves are then used to solve the penalized MVO, producing efficient frontiers and optimal portfolio allocations that reflect each investor's true liquidity preferences. In net return space, the penalized frontier dominates the unpenalized one across all archetypes, with the dominance most pronounced for the pension fund and endowment, where the combination of high payout demand, countercyclical correlation, and binding liquidity constraints generates substantial illiquidity costs at high portfolio illiquidity levels.

8.2 Limitations and future work

Several limitations of the present framework point to natural directions for future research.

The simulation relies on a multivariate normal distribution for asset returns, which abstracts from fat tails, return asymmetries, and the time-varying volatility that characterise real asset markets. Extending the return generation process to accommodate these features, for instance through a regime-switching or stochastic volatility model, would produce more realistic estimates of the illiquidity cost in stress scenarios, where forced selling is most likely and most costly.

A second limitation concerns the liability side of the portfolio. In the present framework, investor heterogeneity is captured through the payout process and the minimum liquidity buffer, which are calibrated separately for each archetype. A natural extension would be to model the asset-liability structure explicitly, distinguishing investors by their duration gap and funding ratio. A pension fund with a large duration gap between assets and liabilities faces a different sensitivity to interest rate movements than one that is well-matched, and this difference feeds directly into the probability of binding liquidity constraints and therefore into the shape of the penalty curve. In this richer setting, the penalty curve would become a function not only of portfolio illiquidity but also of the investor's balance sheet position, providing a more complete and investor-specific characterisation of illiquidity costs.

A third direction for future work concerns the empirical identification of the penalty curve from observed data, rather than from simulation. Given a sufficiently rich panel of institutional portfolios with varying degrees of illiquidity, one could estimate the illiquidity cost curve directly from observed proxies, such as management costs, secondary market transaction records, or realised return gaps between liquid and illiquid portfolios held by the same investor at different points in time. Regressing these cost proxies on portfolio illiquidity levels across investors would yield an econometric estimate of the penalty curve that is grounded in observed behaviour rather than simulated scenarios, providing an independent validation of the simulation-based approach developed in this thesis and a natural benchmark for the calibrated curves.

References

[1] Amundi Investment Institute (2026). *Capital Market Assumptions 2026*. Amundi Asset Management, Paris.

[2] Ang, A., and Bollen, N.P.B. (2010). Locked Up by a Lockup: Valuing Liquidity as a Real Option. *Financial Management*, 39(3), 1069–1096.

[3] Ang, A., Papanikolaou, D., and Westerfield, M.M. (2014). Portfolio Choice with Illiquid Assets. *Management Science*, 60(11), 2737–2761.

[4] Anson, M. (2010). Measuring a Premium for Liquidity Risk. *The Journal of Private Equity*, 13(2), 6–16.

[5] COVIP (2022). *Annual Report 2022*. Commissione di Vigilanza sui Fondi Pensione, Rome.

[6] Derman, E. (2007). A Simple Model for the Expected Premium for Hedge Fund Lockups. *The Journal of Investment Management*, 5(3), 5–15.

[7] Dyck, A., and Pomorski, L. (2011). Is Bigger Better? Size and Performance in Pension Plan Management. *Rotman School of Management Working Paper*. University of Toronto.

[8] Golts, M., and Kritzman, M. (2010). Liquidity Options. *The Journal of Derivatives*, 18(1), 80–89.

[9] Hayes, M., Primbs, J.A., and Chiquoine, B. (2015). A Penalty Cost Approach to Strategic Asset Allocation with Illiquid Asset Classes. *The Journal of Portfolio Management*, 41(2), 33–41.

[10] Hill, J. (2009). A Perspective on Liquidity Risk and Horizon Uncertainty. *The Journal of Portfolio Management*, 35(4), 60–68.

[11] Kinlaw, W., Kritzman, M., and Turkington, D. (2013). Liquidity and Portfolio Choice: A Unified Approach. *The Journal of Portfolio Management*, 39(2), 19–27.

[12] Lee, J. (2012). *Dynamic Portfolio Management with Private Equity Funds*. PhD Thesis, Stanford University, Palo Alto, CA.

[13] Lo, A., Petrov, C., and Wierzbicki, M. (2003). It's 11 p.m.—Do You Know Where Your Liquidity Is? The Mean-Variance-Liquidity Frontier. *The Journal of Investment Management*, 1(1), 55–93.

[14] Markowitz, H. (1952). Portfolio Selection. *Journal of Finance*, 7(1), 77–91.

[15] Mulvey, J., Pauling, W., and Madey, R. (2003). Advantages of Multiperiod Portfolio Models. *The Journal of Portfolio Management*, 29(2), 35–45.

[16] NACUBO–TIAA (2023). *Study of Endowments 2022–2023*. National Association of College and University Business Officers, Washington, DC.

[17] Preqin (2025). *Institutional Allocation Study 2025*. Preqin Ltd., London.

[18] Takahashi, D., and Alexander, S. (2002). Illiquid Alternative Asset Fund Modeling. *The Journal of Portfolio Management*, 28(2), 90–100.

[19] Terhaar, K., Staub, R., and Singer, B. (2003). Appropriate Policy Allocation for Alternative Investments. *The Journal of Portfolio Management*, 29(3), 101–110.

Annex A - Parameters Sensitivity

This annex reports the sensitivity of optimal portfolio composition to the choice of penalty function parameters. For each of the three functional forms considered in Section 4, the penalized mean-variance optimization is solved across a 6×6 grid of (α, β) values, and the resulting portfolio composition along the efficient frontier is displayed as a stacked area chart. Rows correspond to increasing values of α and columns to increasing values of β. The analysis is intended to provide the practitioner with an intuitive map of how the penalty parameters translate into portfolio allocations, independently of the calibration methodology developed in Section 6.

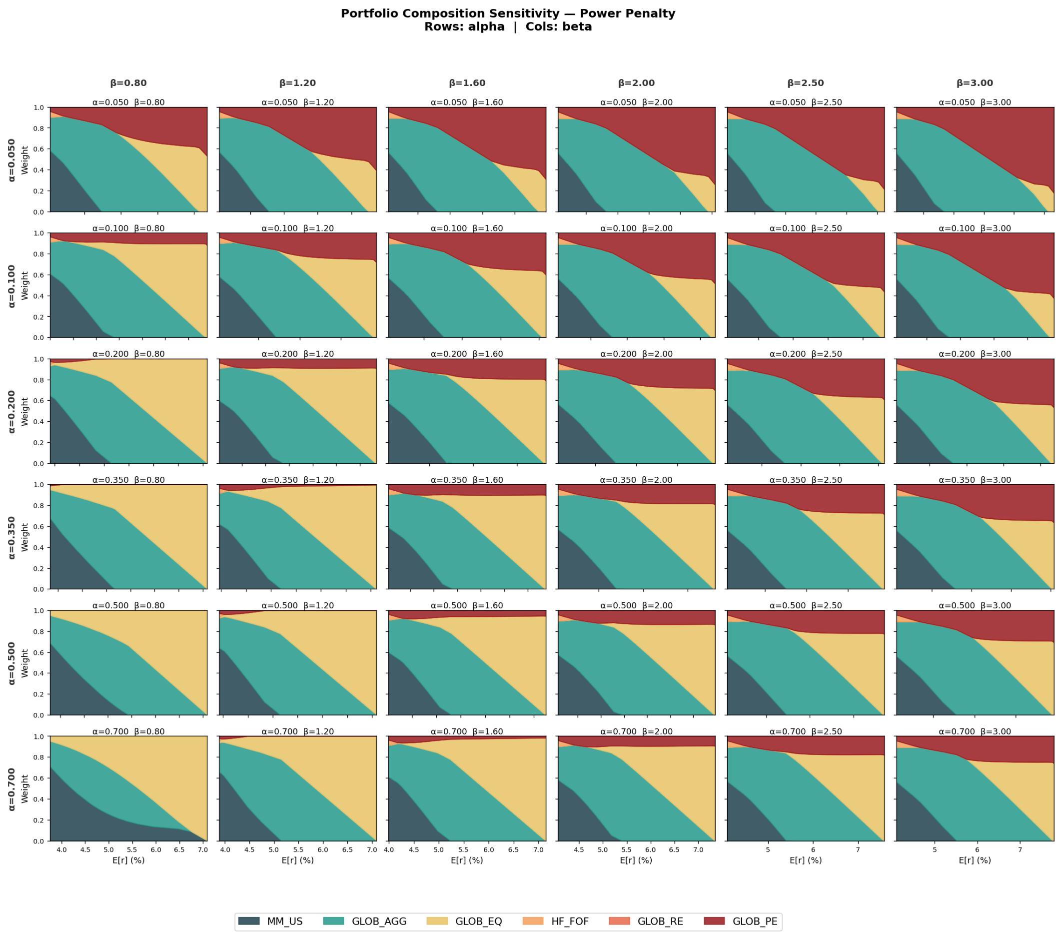

Figure 17: Portfolio composition sensitivity matrix for the power penalty $c(L) = \alpha \cdot L^{\beta}$. Rows: $\alpha \in \{0.05, 0.10, 0.20, 0.35, 0.50, 0.70\}$. Columns: $\beta \in \{0.80, 1.20, 1.60, 2.00, 2.50, 3.00\}$.

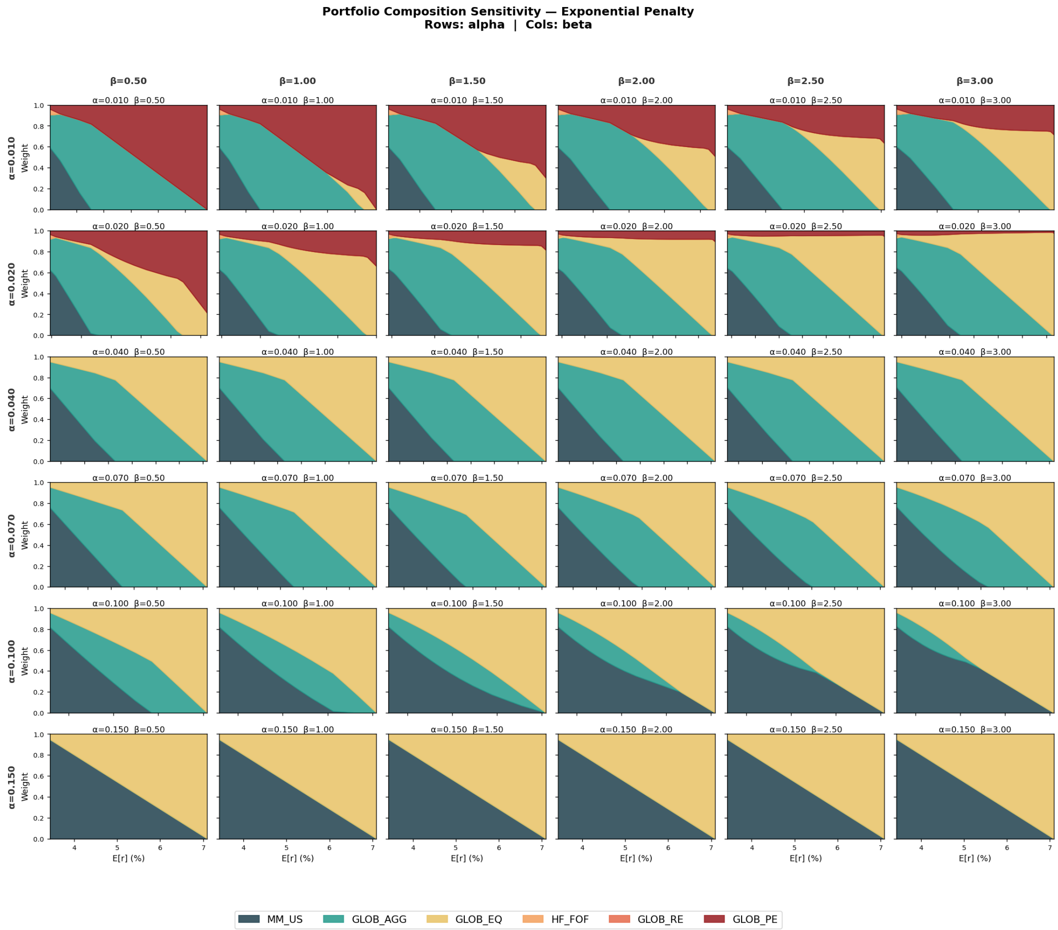

Figure 18: Portfolio composition sensitivity matrix for the exponential penalty $c(L) = \alpha \cdot e^{\beta L}$. Rows: $\alpha \in \{0.01, 0.02, 0.04, 0.07, 0.10, 0.15\}$. Columns: $\beta \in \{0.50, 1.00, 1.50, 2.00, 2.50, 3.00\}$.

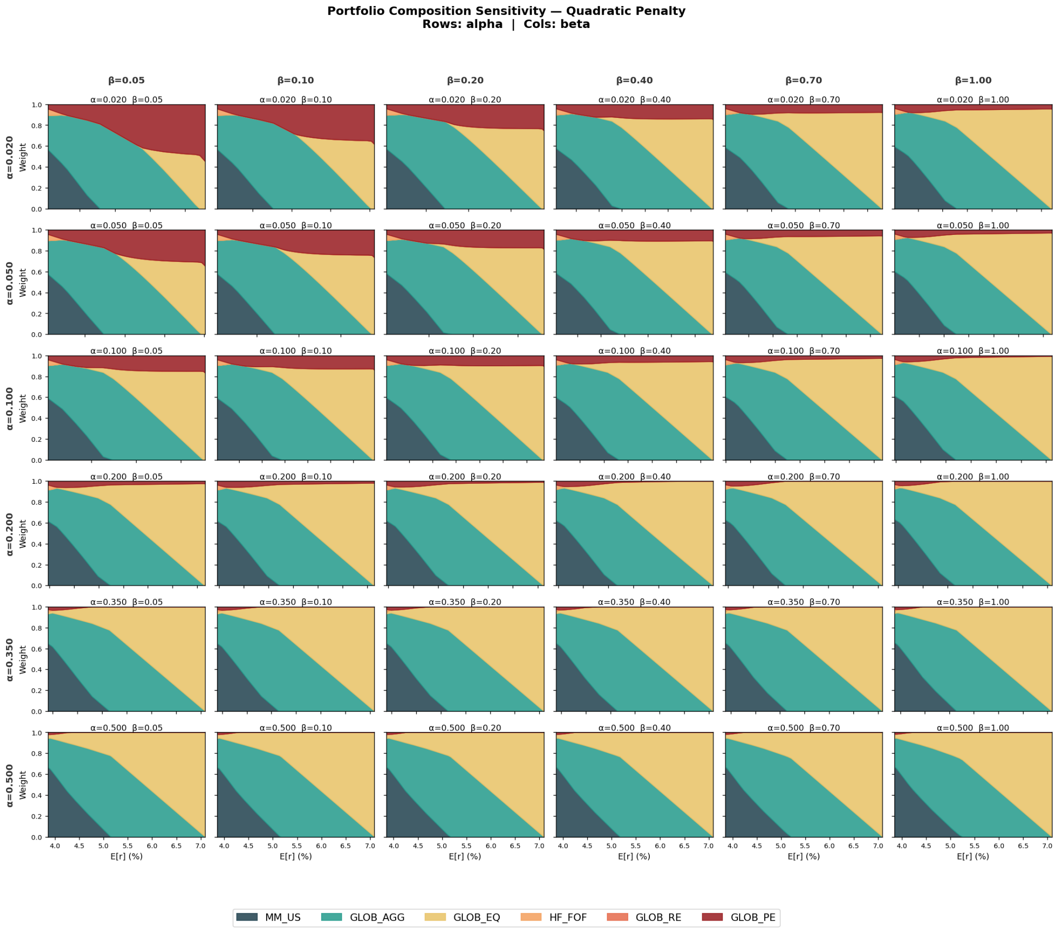

Figure 19: Portfolio composition sensitivity matrix for the quadratic penalty $c(L) = \alpha L + \beta L^2$. Rows: $\alpha \in \{0.02, 0.05, 0.10, 0.20, 0.35, 0.50\}$. Columns: $\beta \in \{0.05, 0.10, 0.20, 0.40, 0.70, 1.00\}$.